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SURGICAL AI
COLLABORATIVE



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Surgical Data Science: Promises, Perils and Realistic Expectations

Amin Madani, MD PhD FRCSC FACS

Endocrine/Acute Care Surgery, University Health Network

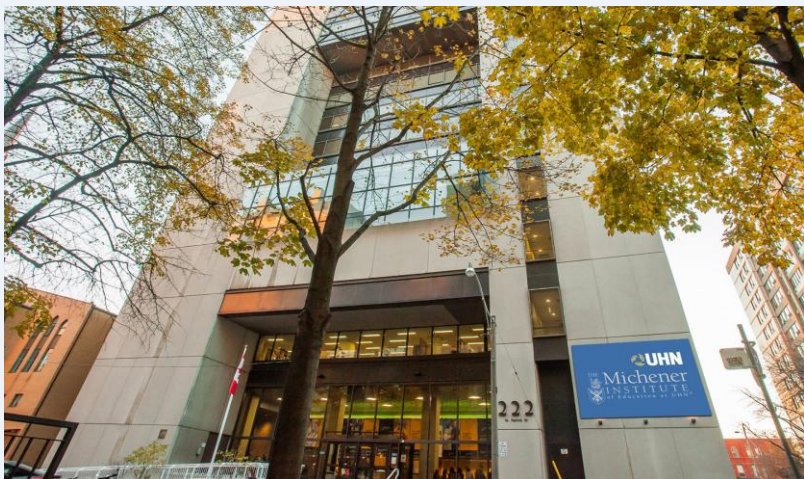
Assistant Professor of Surgery, University of Toronto

Director, Surgical Artificial Intelligence Research Academy

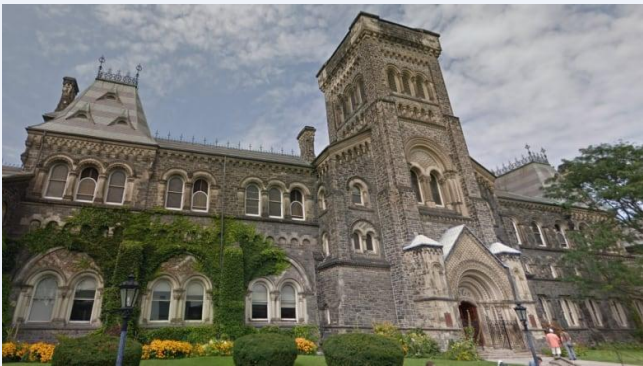
Co-Founder and Chair, Global Surgical Artificial Intelligence Collaborative



@aminmadaniMD @surgicalAI_Lab



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My Lab



Multidisciplinary:

- Engineers
- Computer scientists
- Data scientists
- Education psychologist
- Game developers
- Software developers



My Lab

Education

Technology

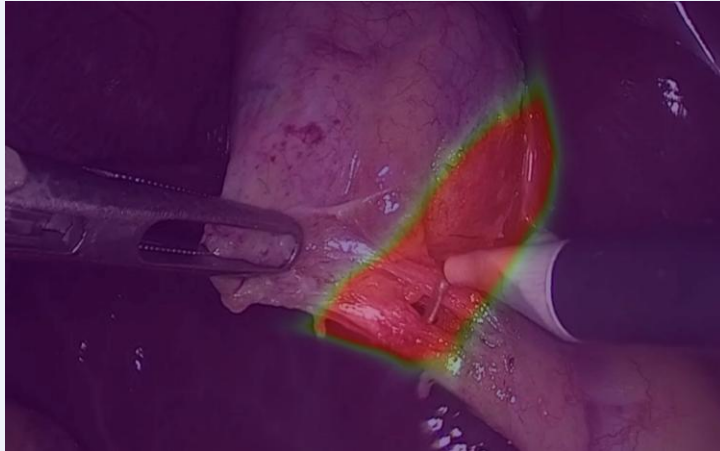


Innovation

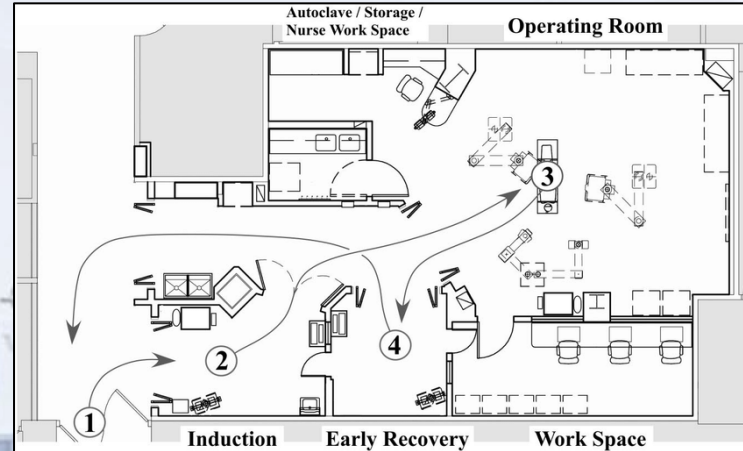
Patient Safety

Surgical Data Science

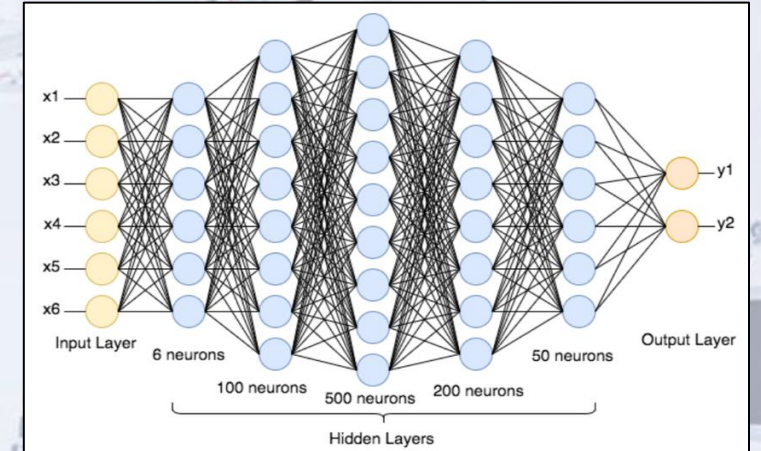
Real-time guidance



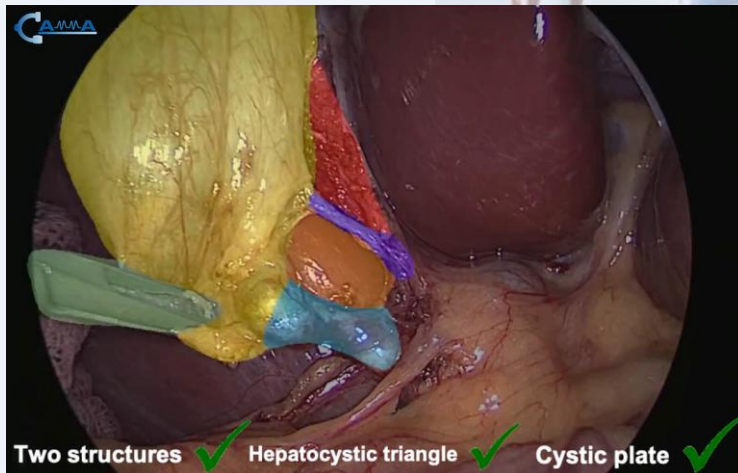
Patient flow/resource management



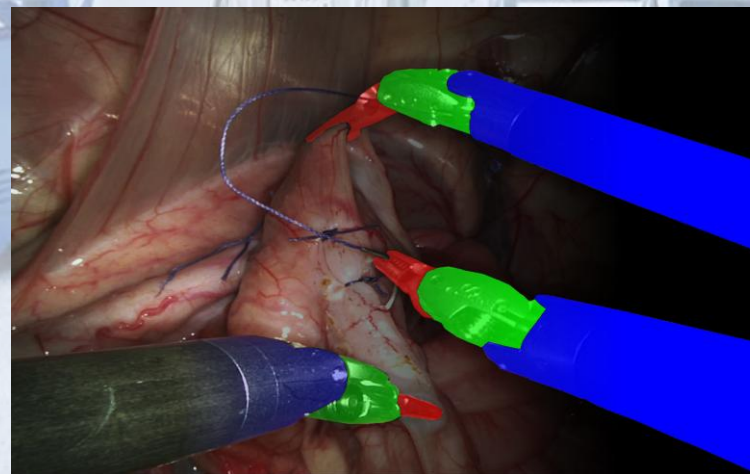
Perioperative care



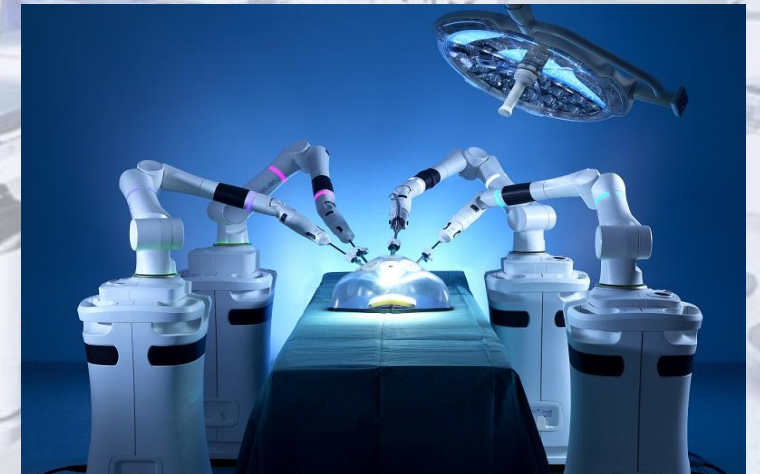
Quality-improvement



Surgical education

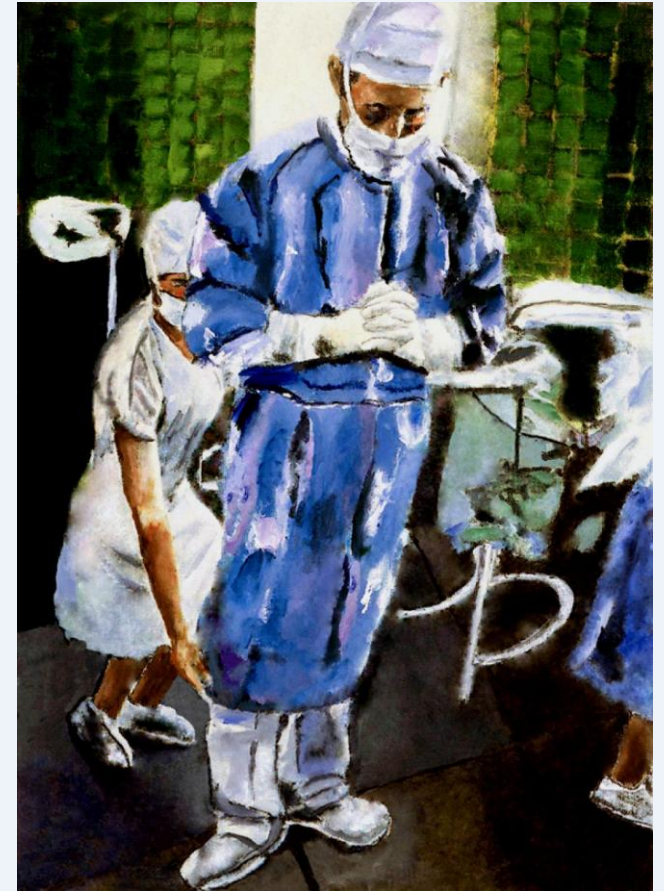


Autonomous surgery



Hospitals are Dangerous Places

- High rates of adverse events
- Surgical patients – 3-5% suffer complications
 - >75% preventable
 - “Technique-related” complication
- WHO: 200-300 million operations annually
 - >3 million major preventable complications annually
 - >500,000 preventable peri-operative deaths



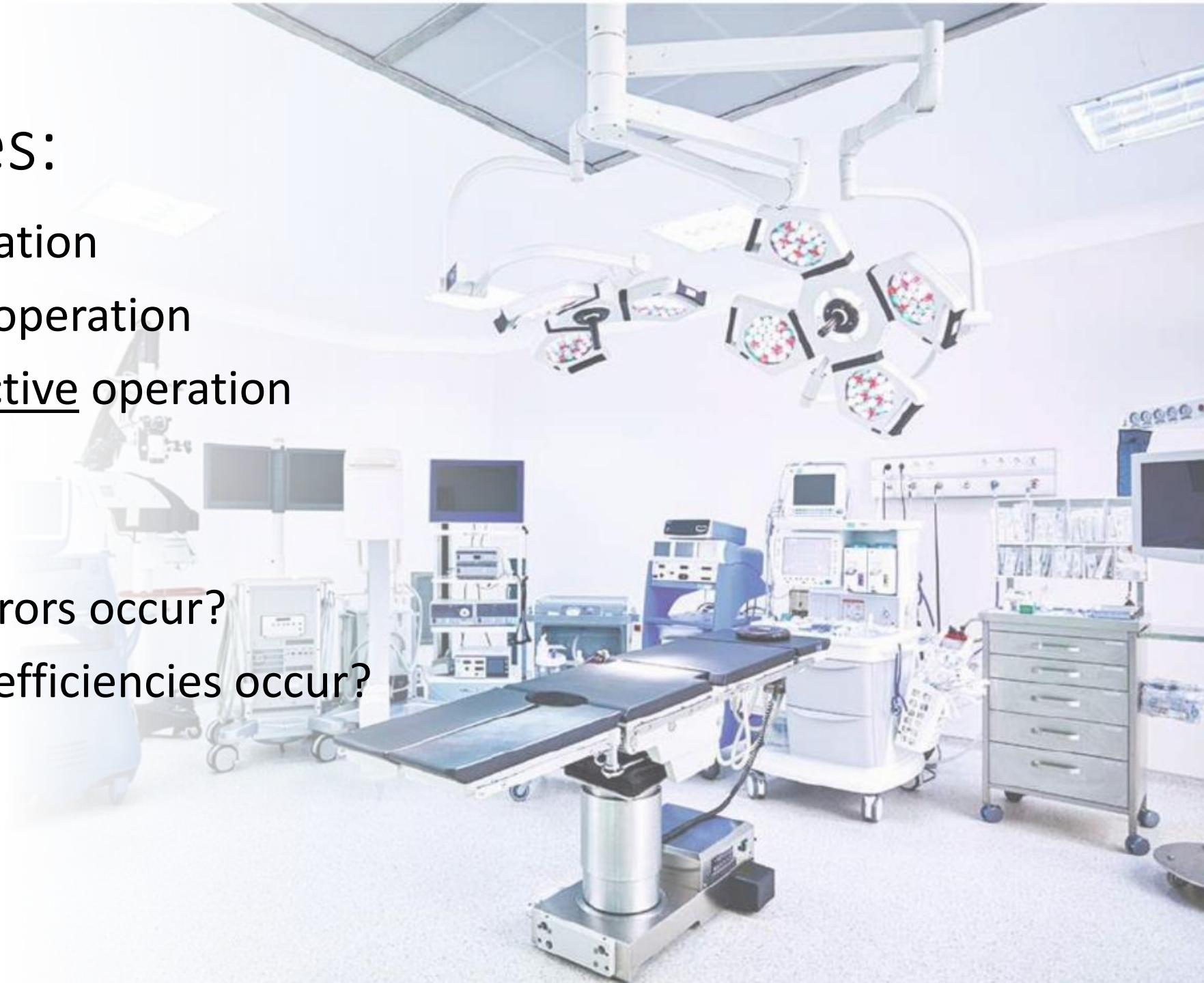
Rogers SO, et al. Surgery. 2006.
Gawande AA, et al. Surgery 1999.

OR Objectives:

- Perform safe operation
- Perform effective operation
- Perform cost-effective operation

Unmet Needs:

- Where/How do errors occur?
- Where/How do inefficiencies occur?



Surgical Performance

“An operation calls for 70-75% decision-making and 20-25% dexterity”

– Frank Spencer



Rogers SO, et al. Surgery. 2006.
Way LW, et al. Annals of Surgery. 2003.

Judgment



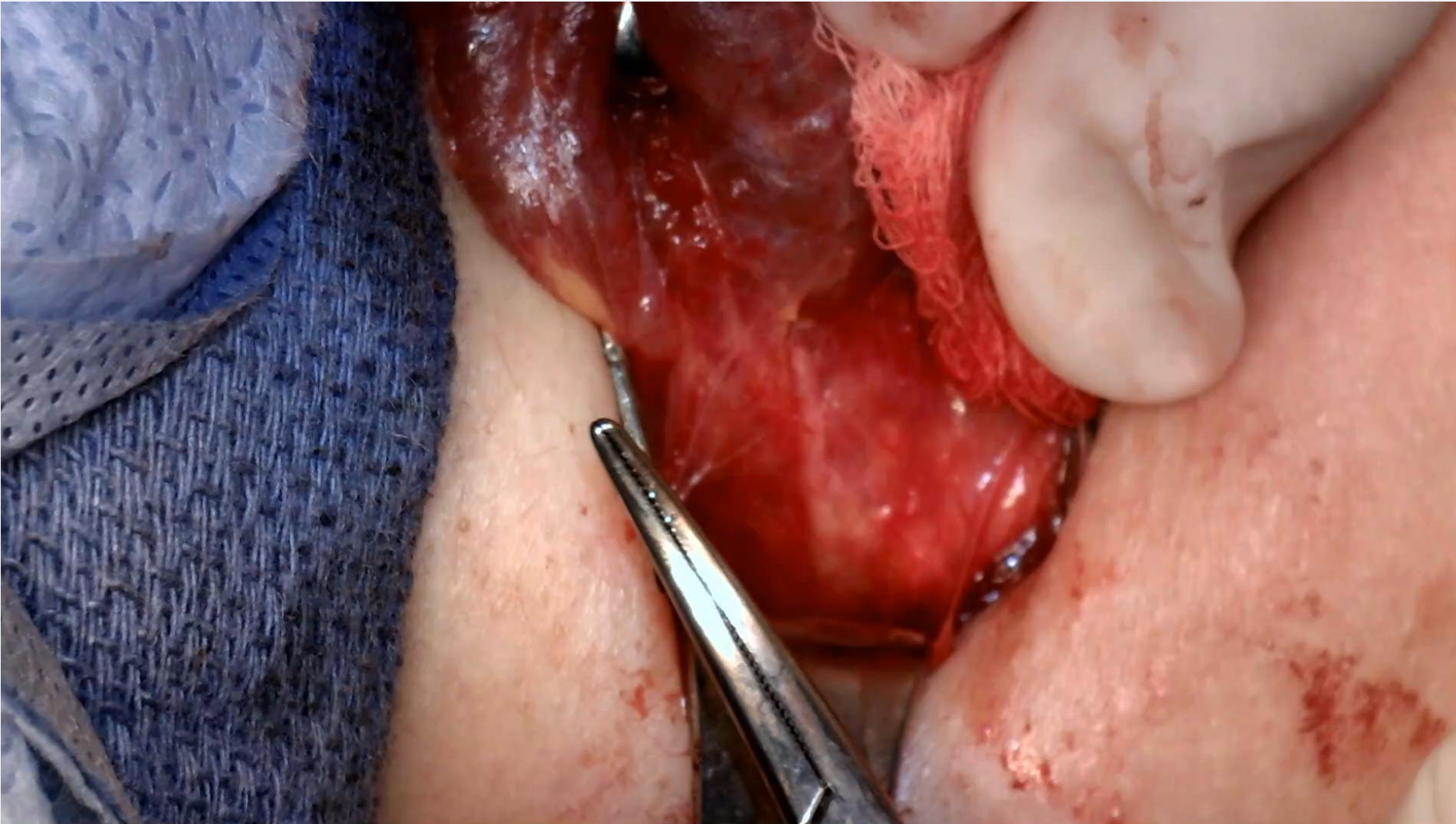
Performance



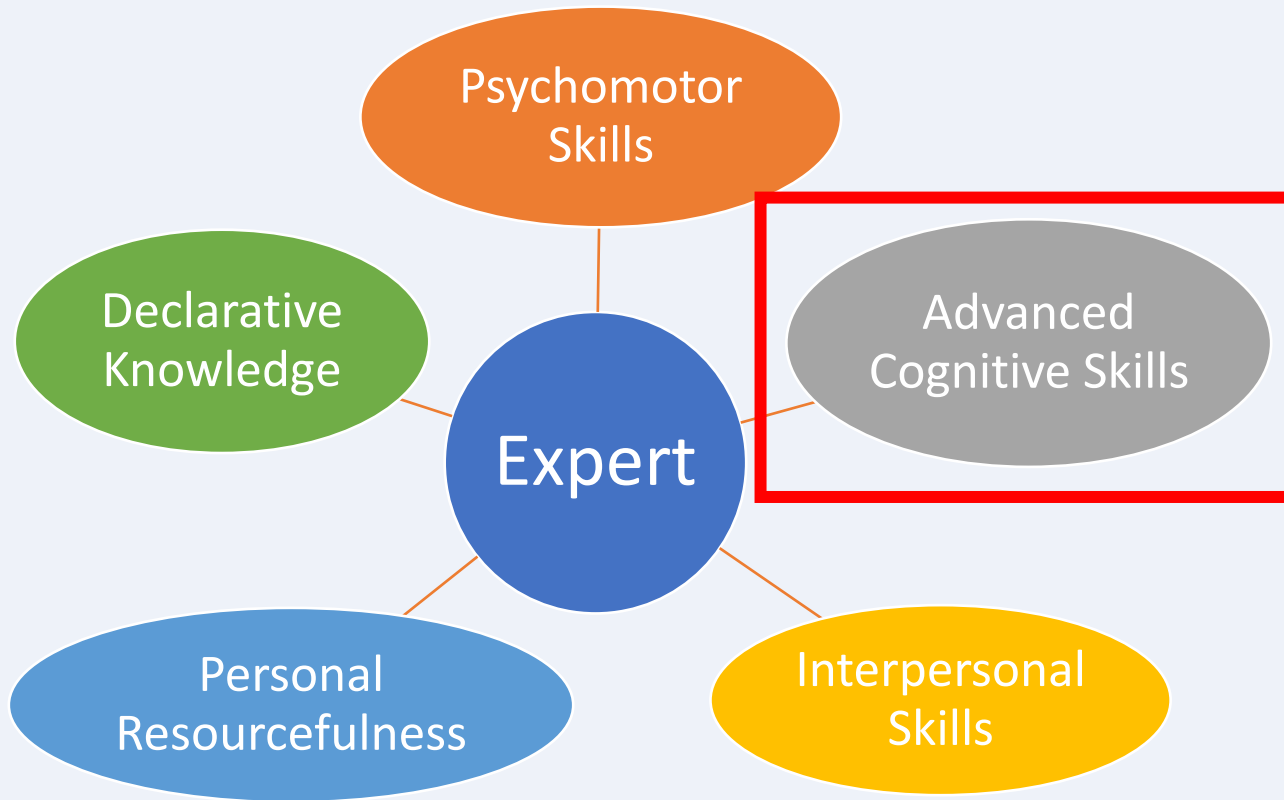
Outcomes

Thyroidectomy

- Recurrent laryngeal nerve dissection



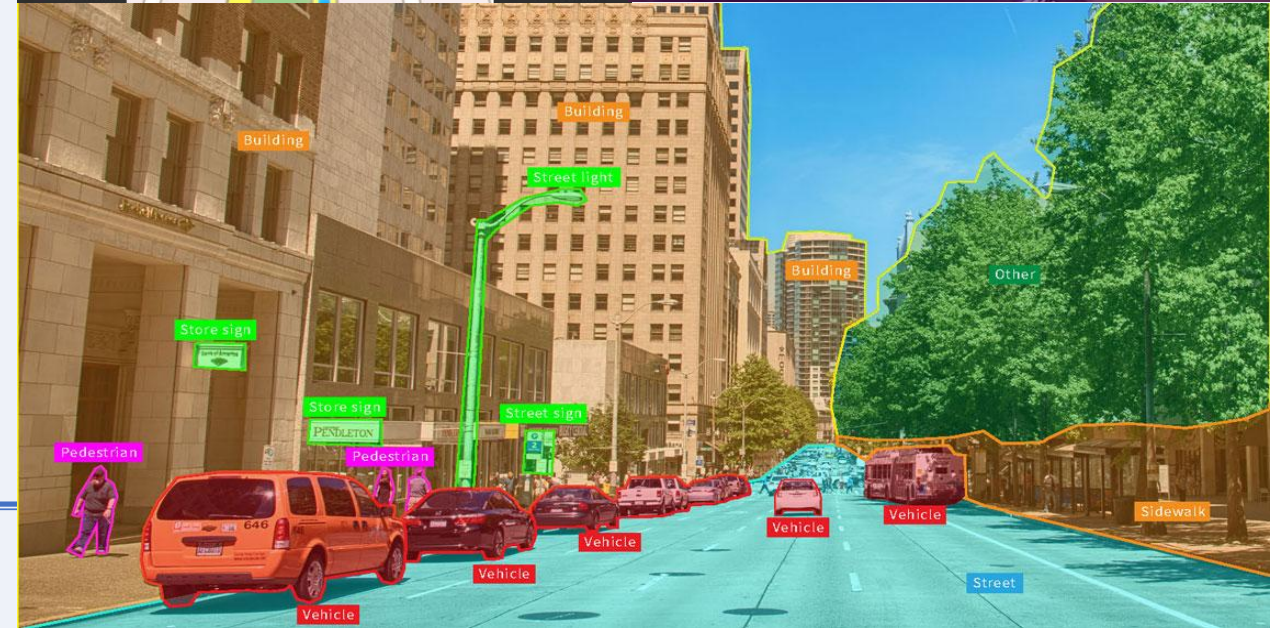
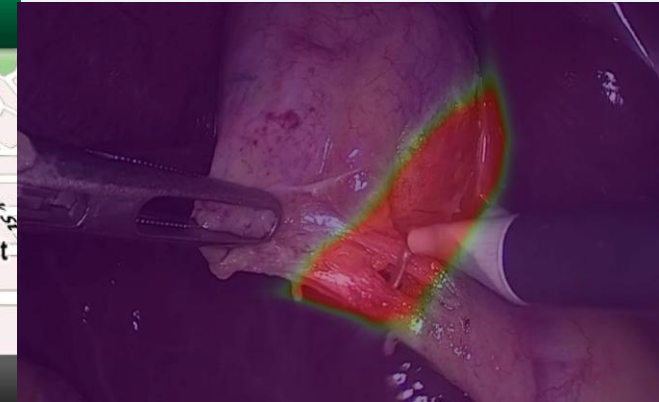
Surgical Performance



Madani A, et al. Annals of Surgery. 2016;265:255-267.

Surgical Artificial Intelligence

- Provides the means for perception and cognition
- Computer vision:
 - Deep learning to cluster pixels and extract high-level features
 - Achieve human-level recognition of objects and patterns
 - E.g. object tracking, scene recognition



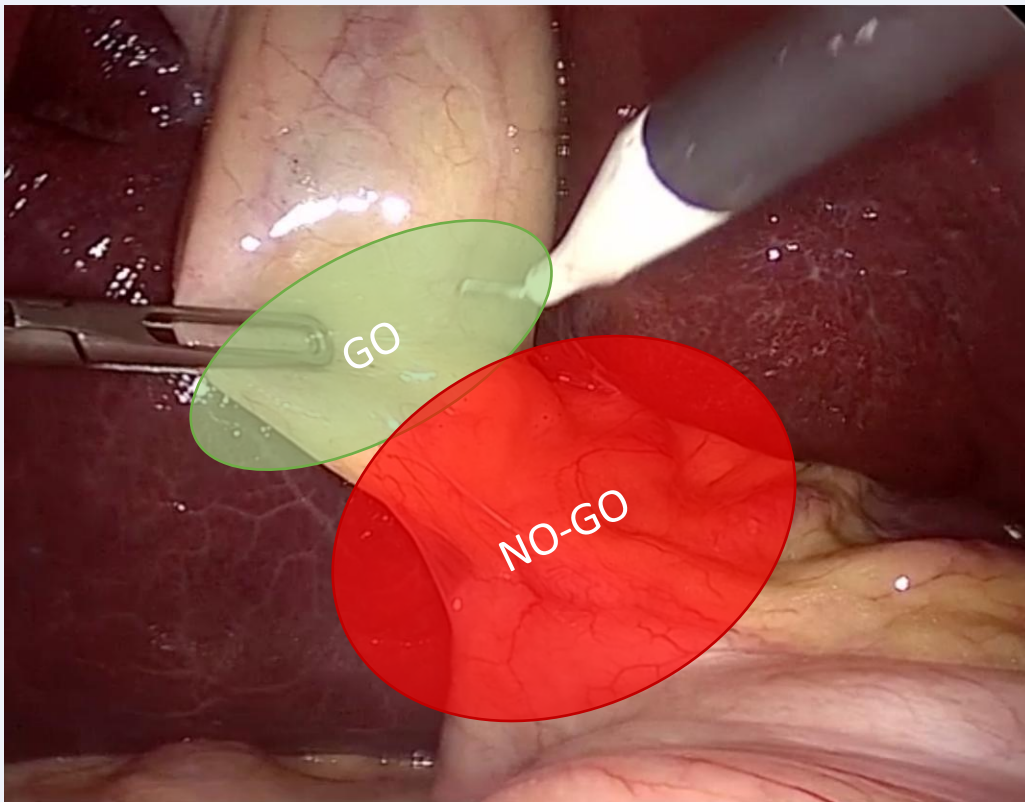


A close-up of a man's face with a surprised expression, overlaid with various futuristic digital and technical graphics. These include glowing red and blue lines, circular patterns, a grid of small circles near the mouth, and various alphanumeric codes and symbols scattered across the image. The overall aesthetic is high-tech and digital.

Use-Case 1: Surgical AI for Performance Augmentation

Surgical Artificial Intelligence

- GO and NO-GO zones
- Dataset: 2624 images from 308 videos

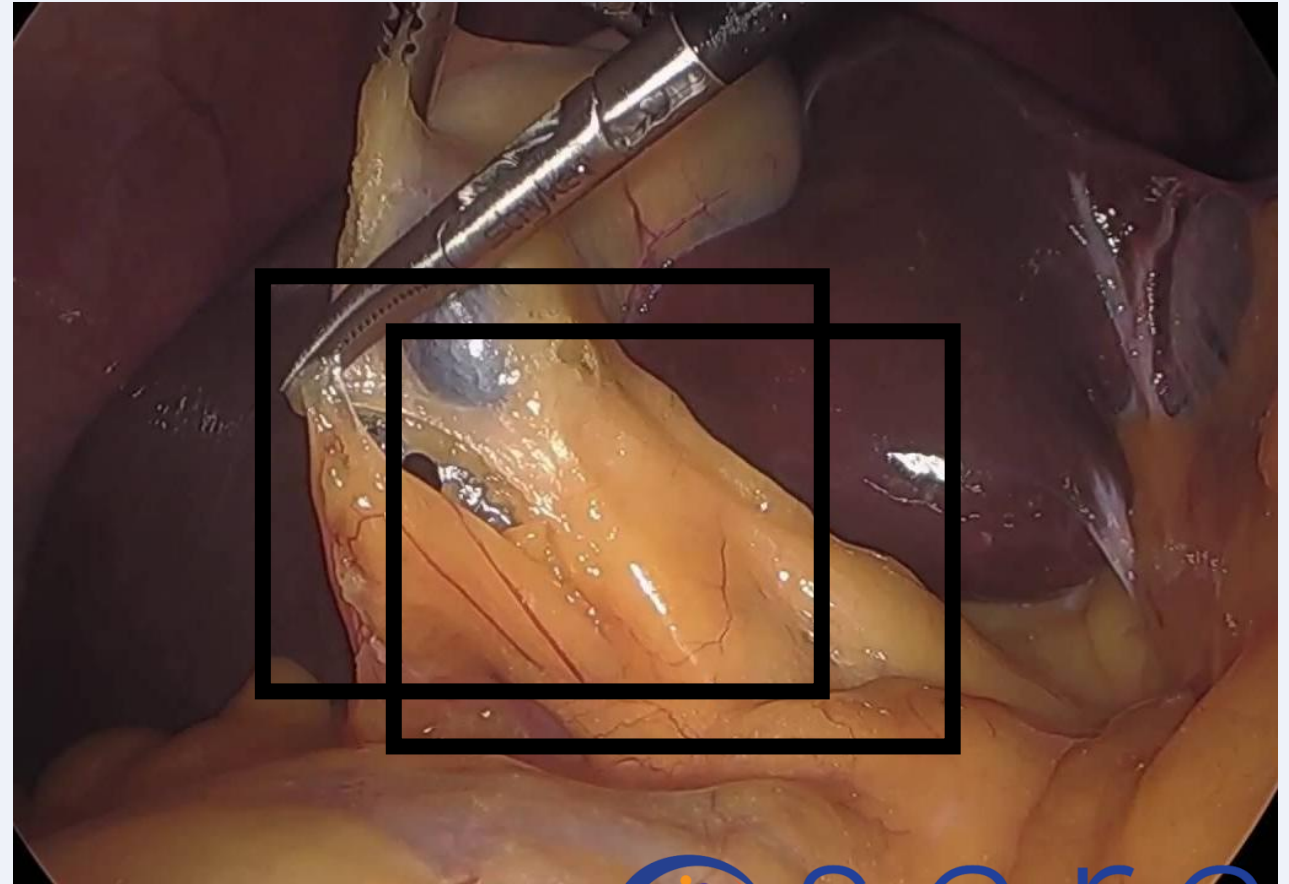


GO ZONE = SAFE
NO-GO ZONE = DANGER

Computer Vision in the Surgical Field

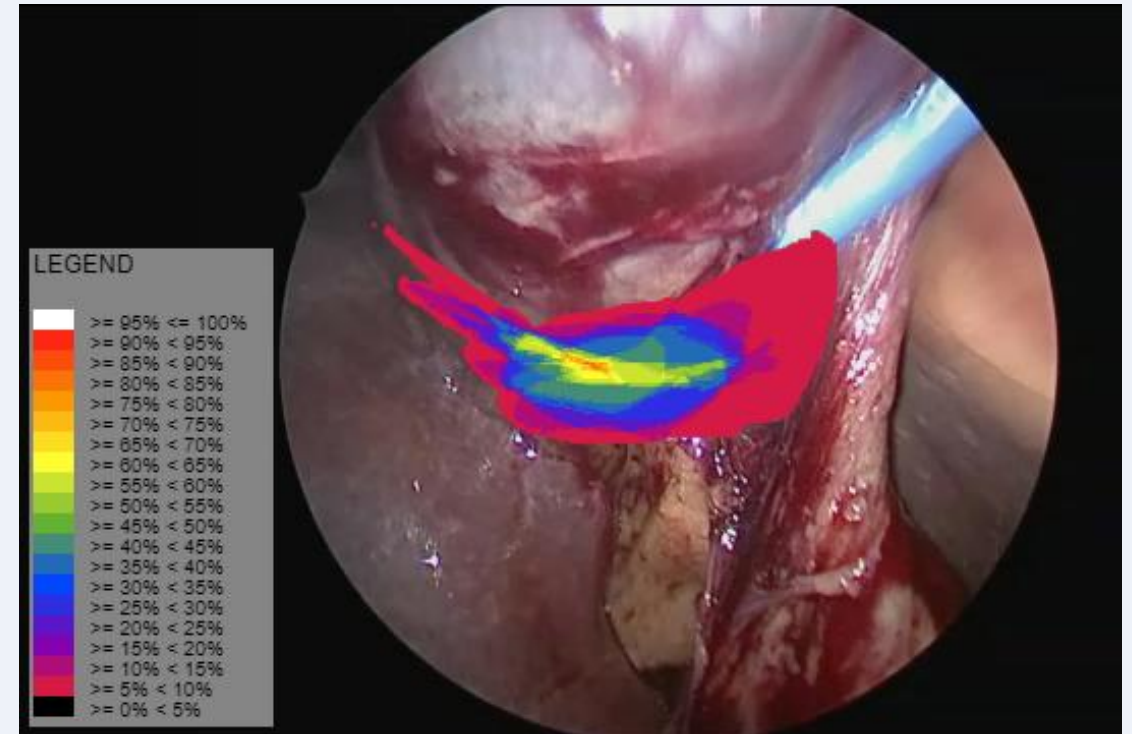
Challenges:

- Structures almost always not immediately obvious
 - Fatty and fibrous tissues
- Bounding boxes have no role
- Imperfect inter-expert agreement requires consensus
- Limited data



Annotation Methodology

- Visual concordance test
- Gold standard reference
- Panel of experts' consensus



Madani A, et al. BJS. 2019.

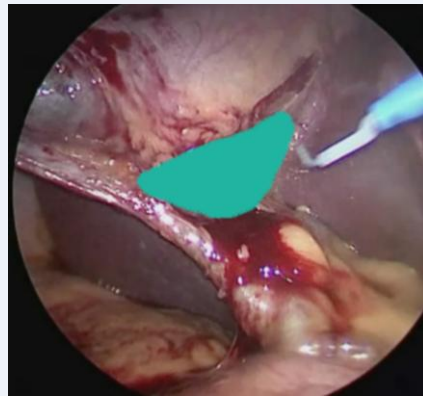
Madani A, et al. JAMA Surgery. 2020.

Surgical Artificial Intelligence

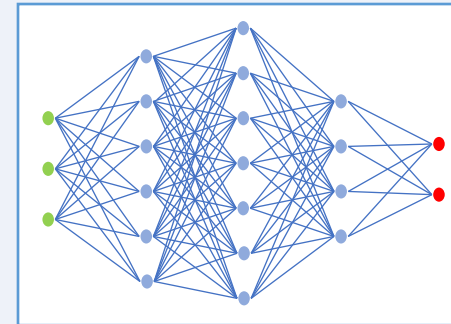
- Training of Deep Neural Networks



Expert Surgeon Panel



Annotations
("free-hand drawings")



Deep Neural Network



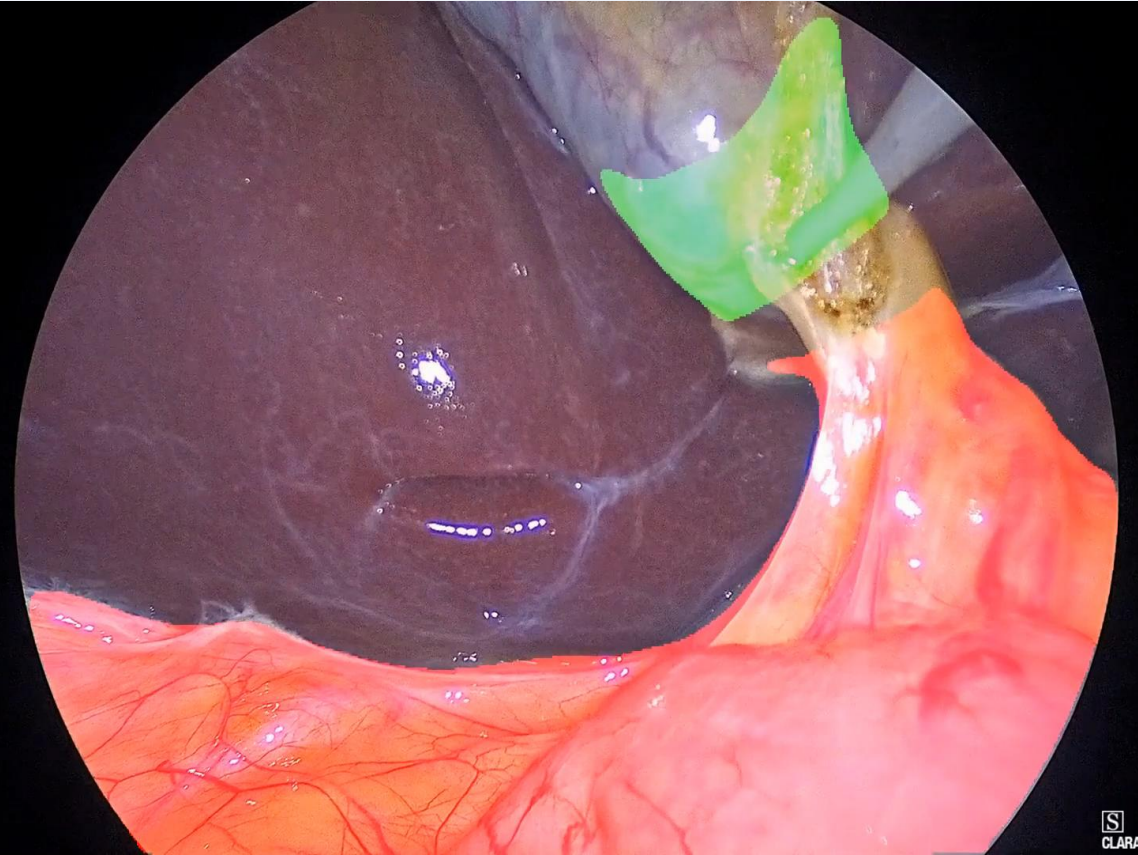
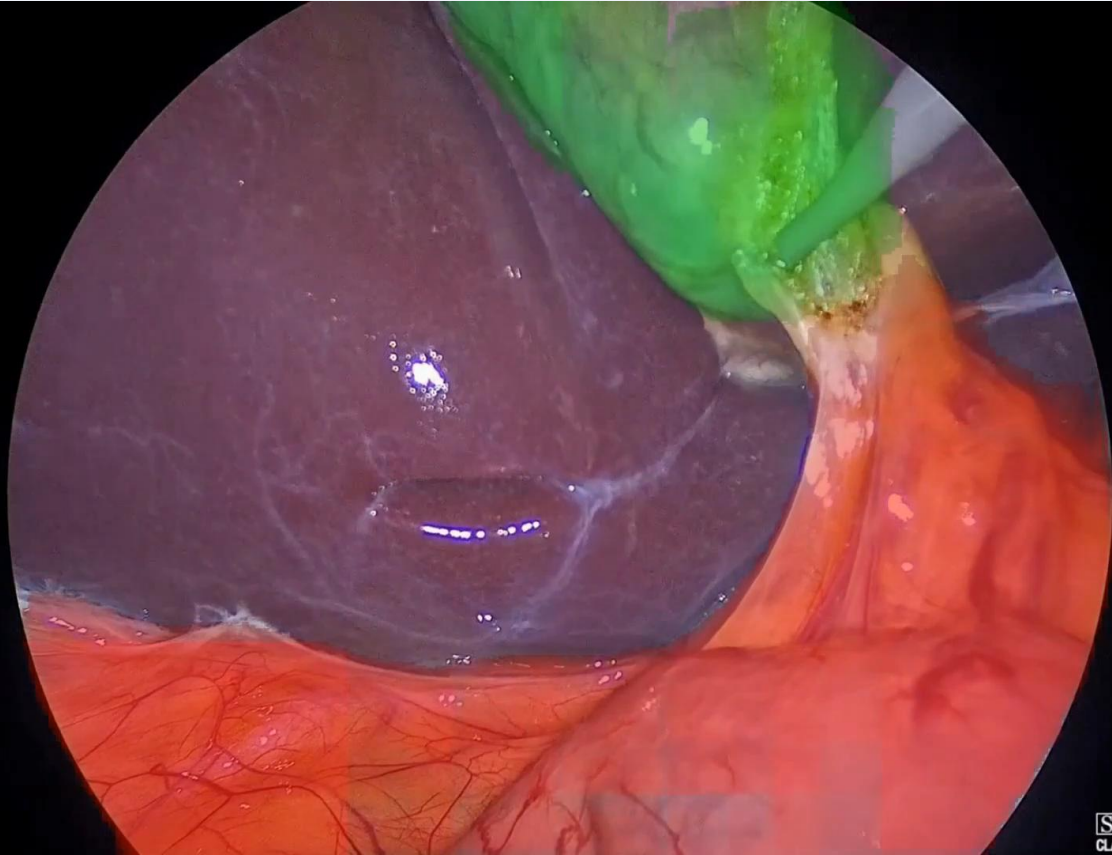
**AI Model Capable of
Predicting Location of
Target Structures**

Annotation Protocol

Checklist

- Data source
- Data sampling for annotation
- Define CV task
- Annotation process
- Annotator selection
- Annotator variability – consensus
- Quality assurance check

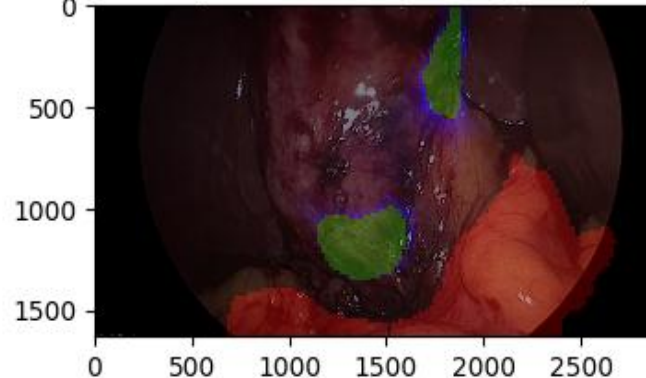




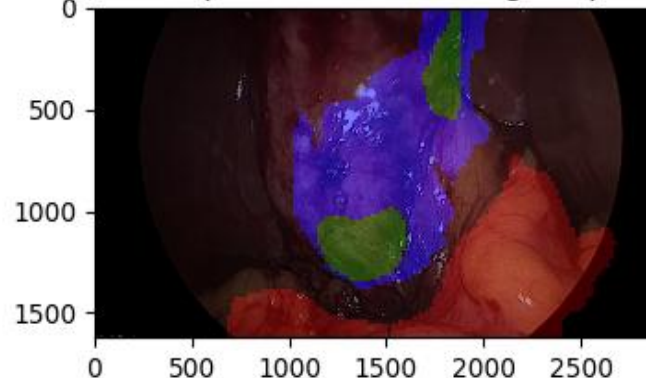
GoN-GM

- Validation
- Testing
- Model
- Self-supervised annotation

probability added pixels outside of original prediction space



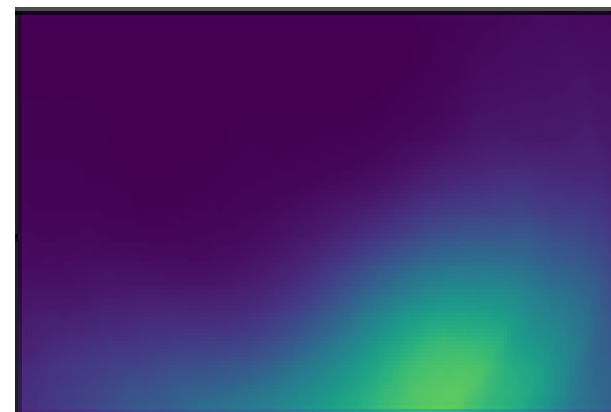
Range of all added pixels outside of original prediction space



GO Zone

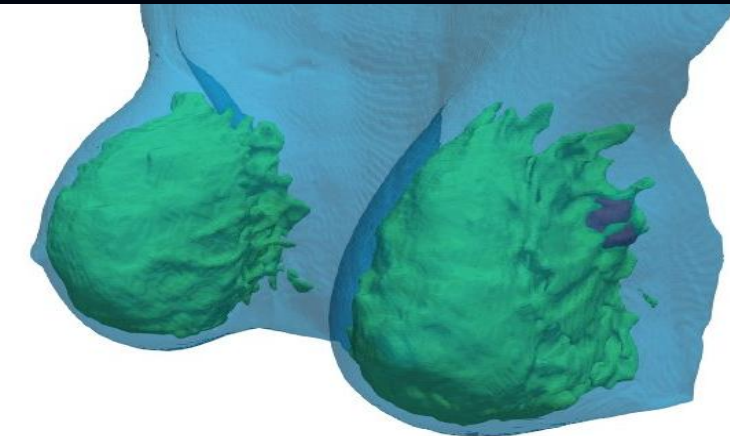
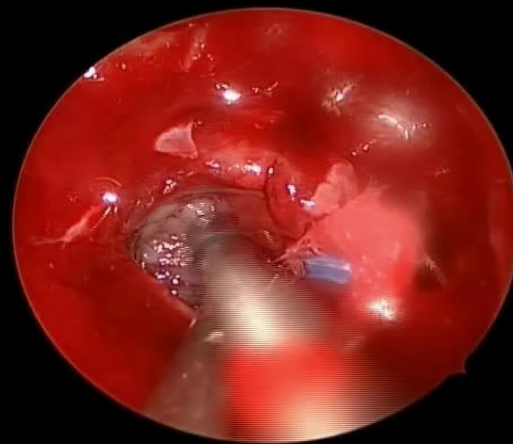
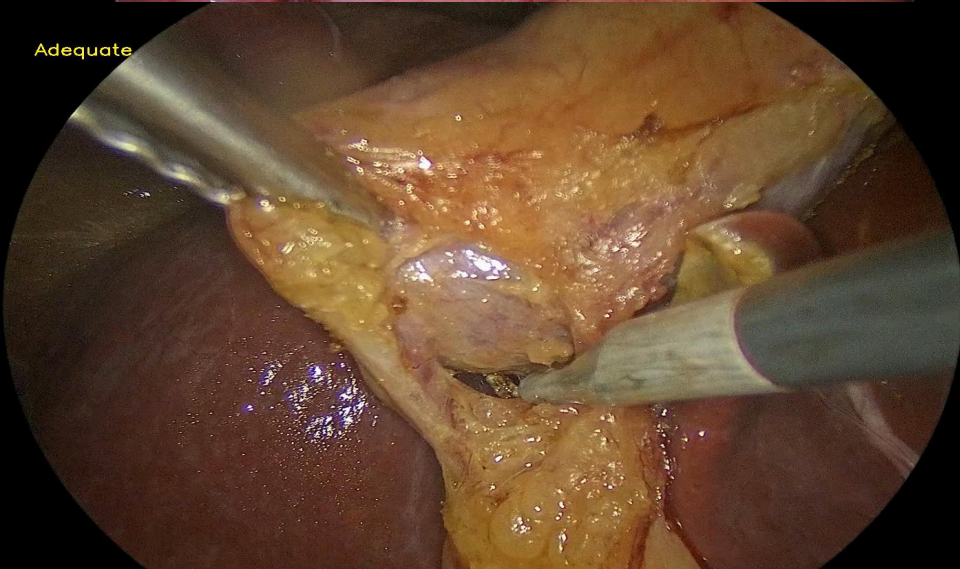
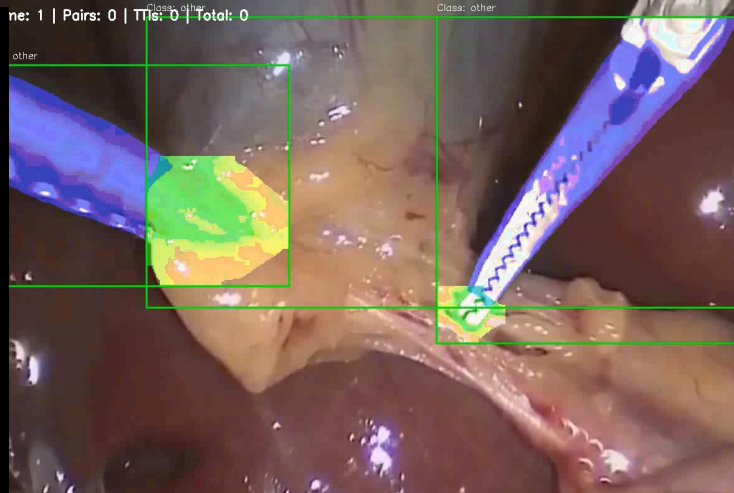
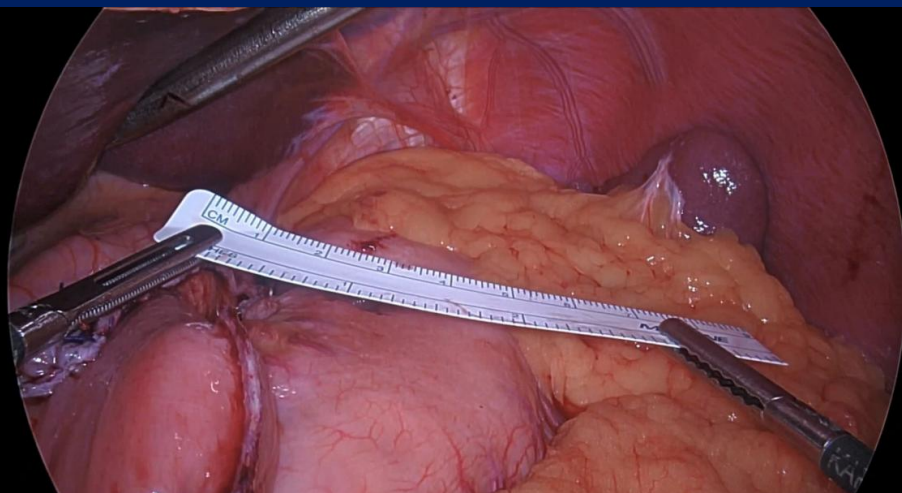


NO-GO Zone

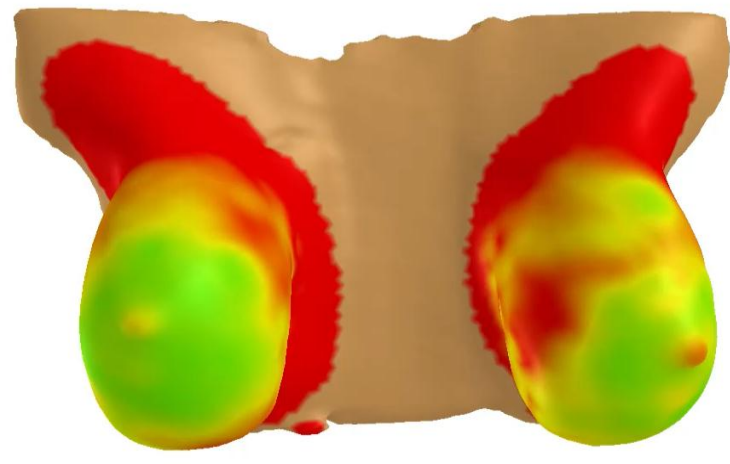


Laplante S, et al. Surgical Endoscopy 2022.

Images courtesy of: Jaryd Hunter, Pouria Mashouri, Michael Brudno

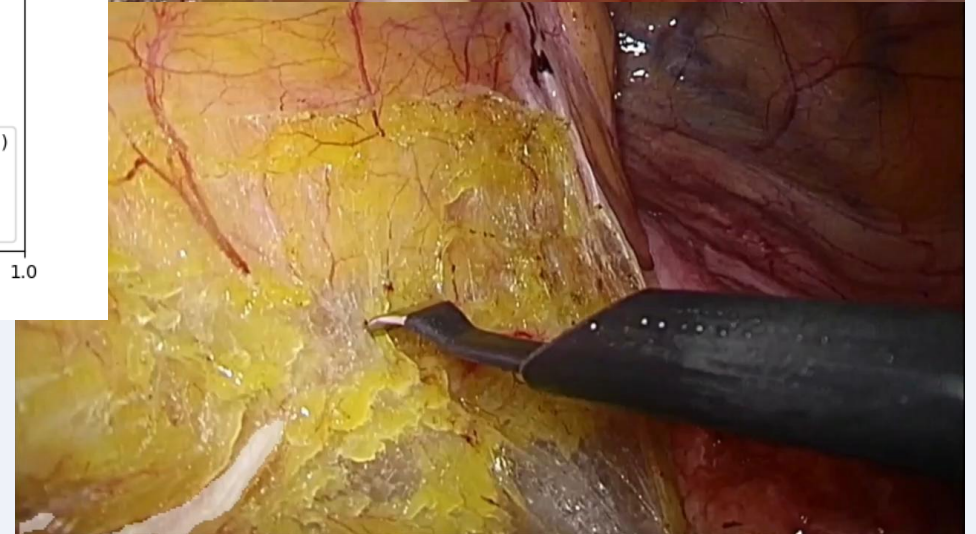
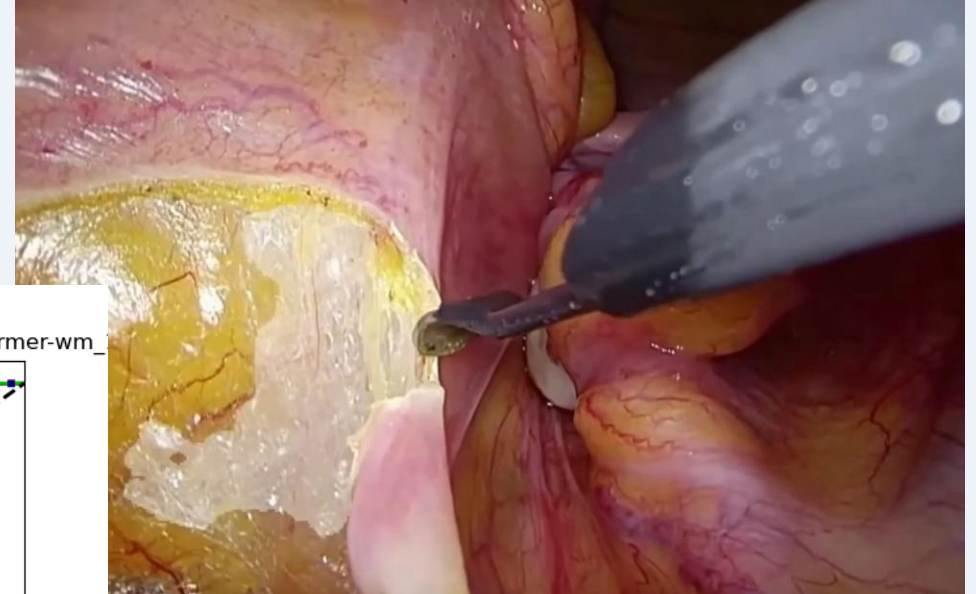
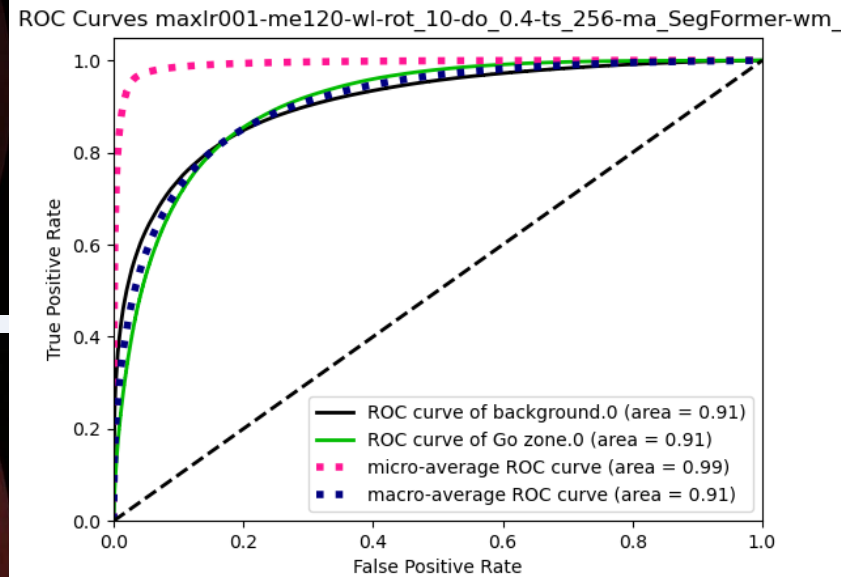
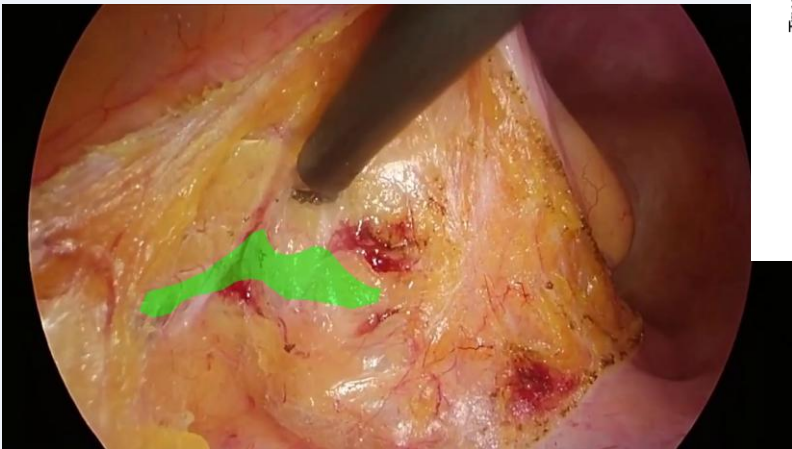
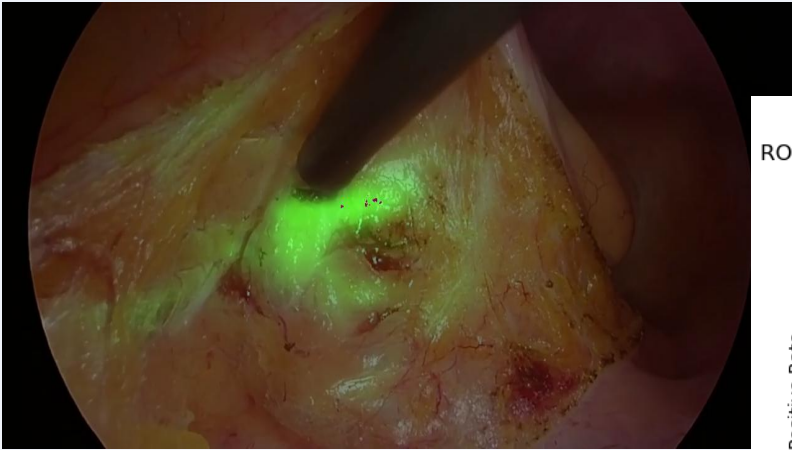


CLARA



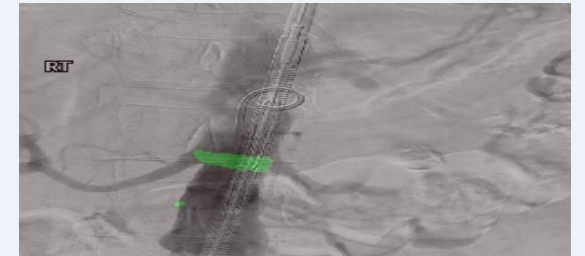
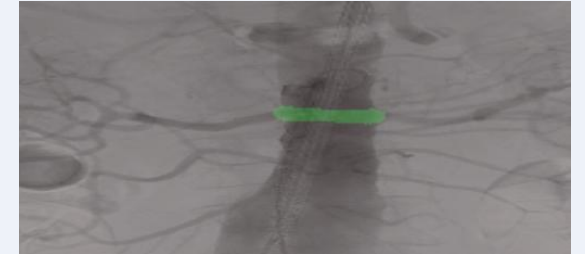
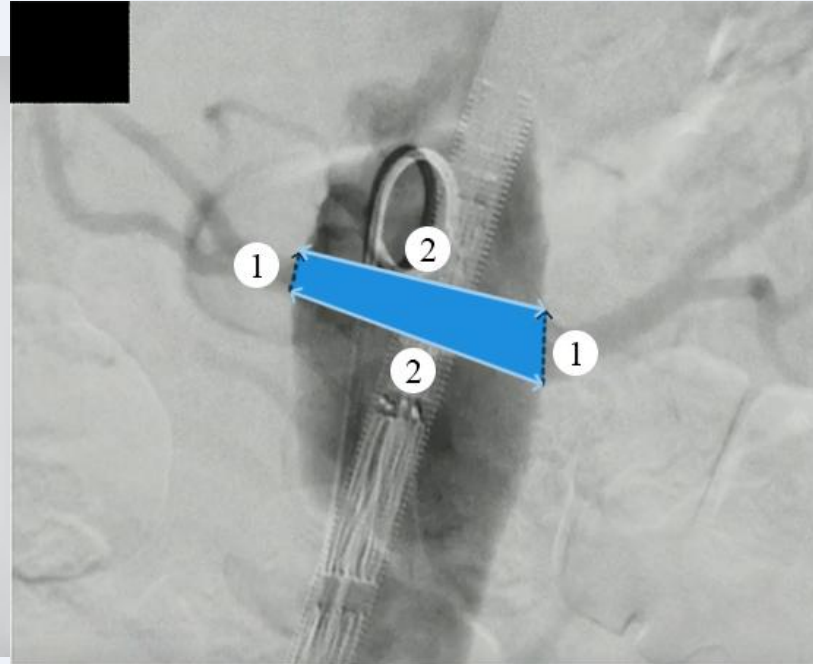
Computer Vision in the Surgical Field

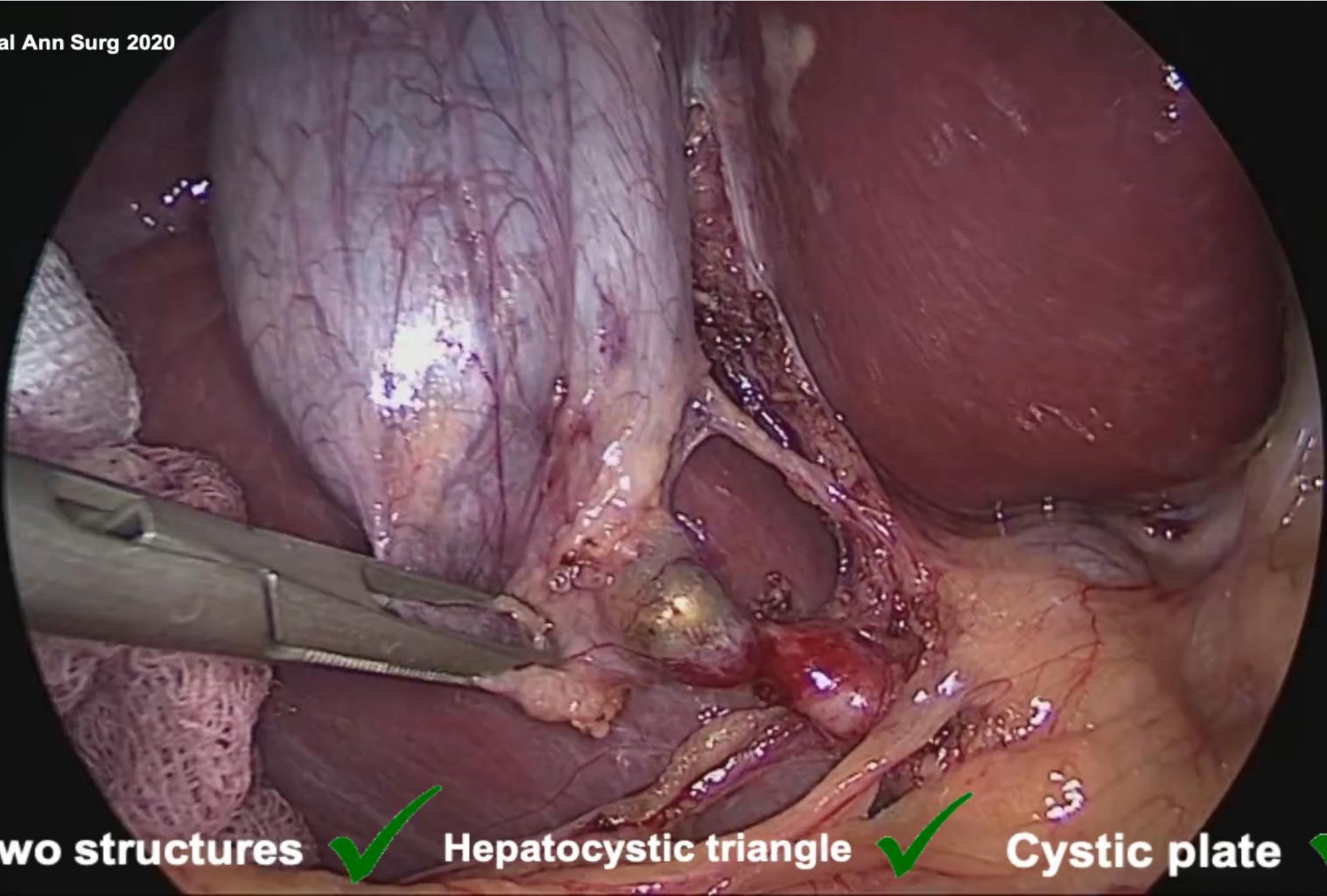
- Colorectal surgery



Computer Vision in the Surgical Field

- No-Go zone graft deployment





Two structures ✓

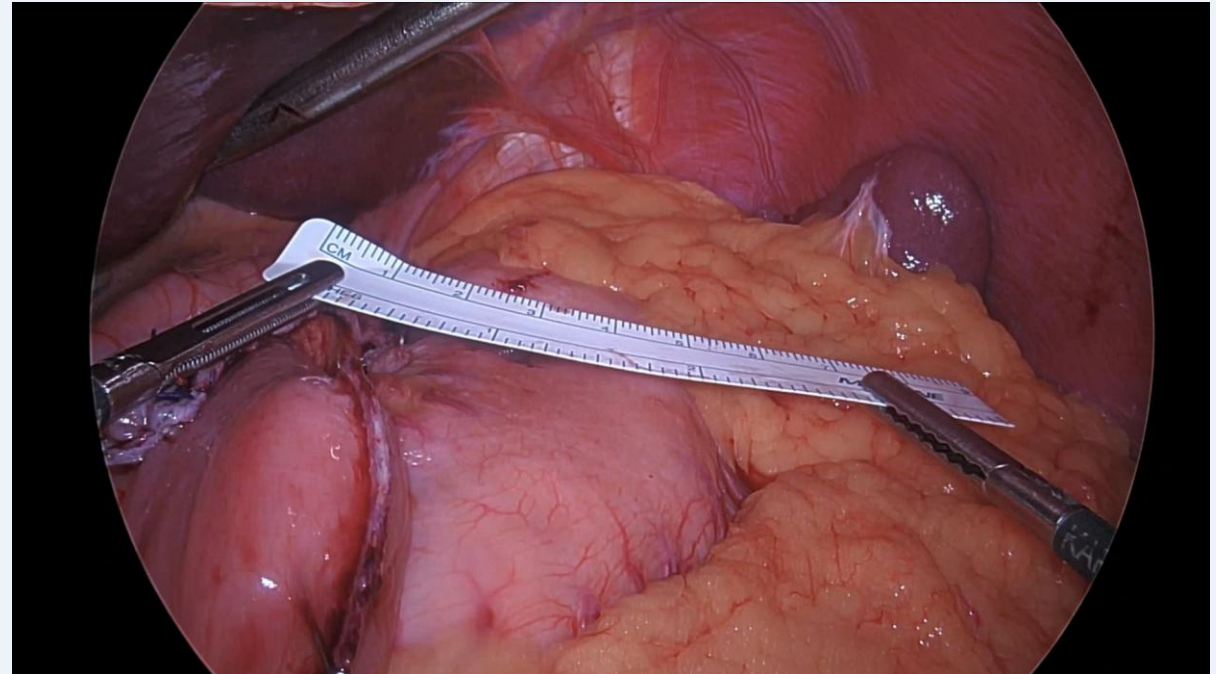
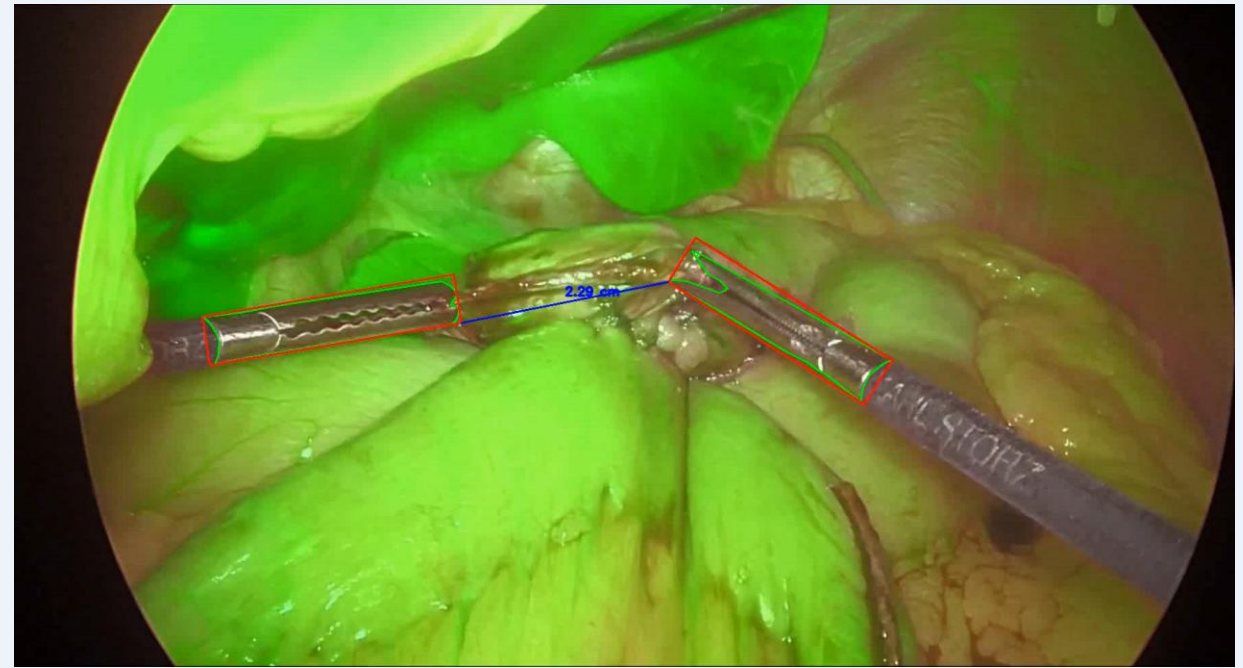
Hepatocystic triangle ✓

Cystic plate ✓

Digital Ruler

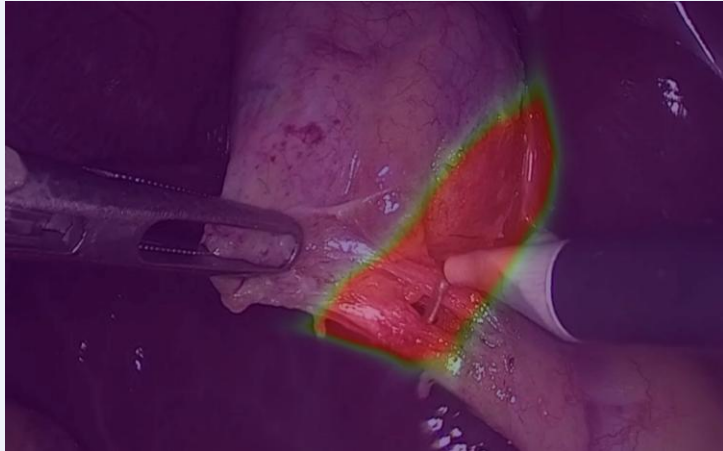
Challenges:

- Measure intracorporeal distances
 - E.g. length of bowel during obesity surgery (bypass)
 - E.g. dimensions of hernia defect

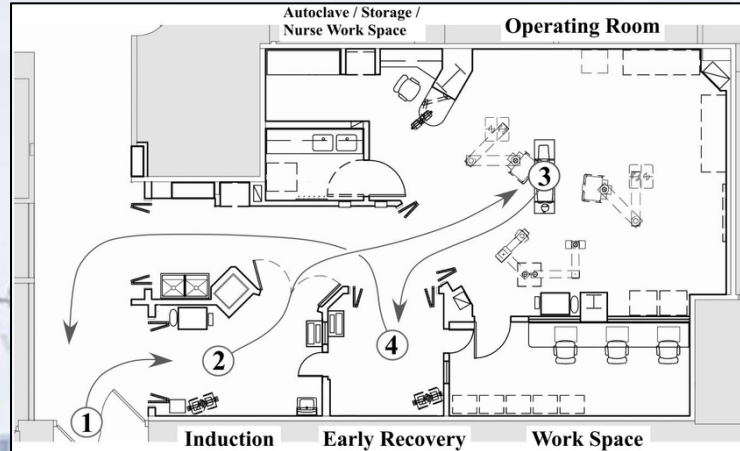


Surgical Data Science

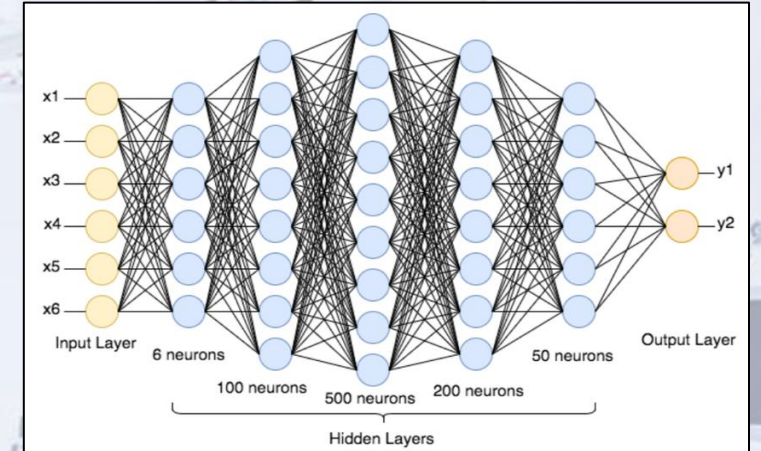
Real-time guidance



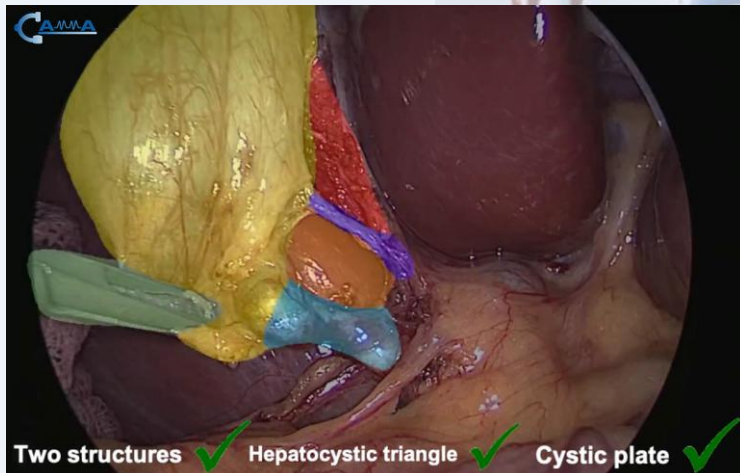
Patient flow/resource management



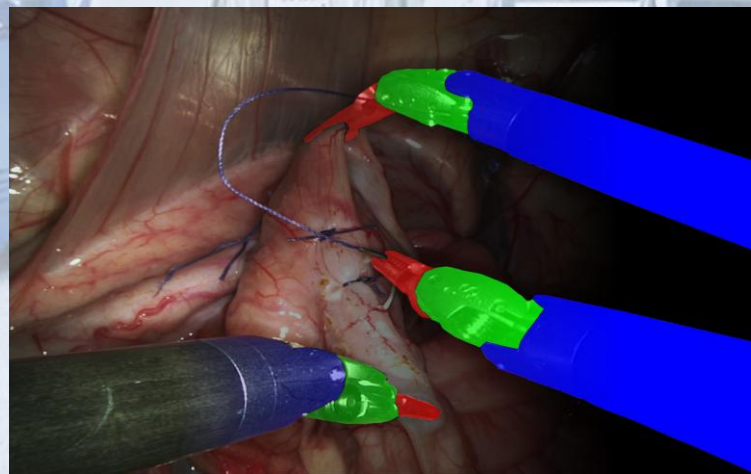
Perioperative care



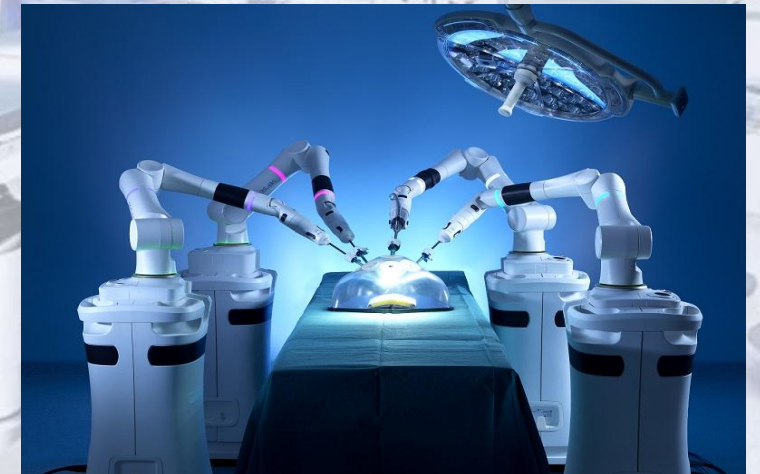
Quality-improvement



Surgical education



Autonomous surgery



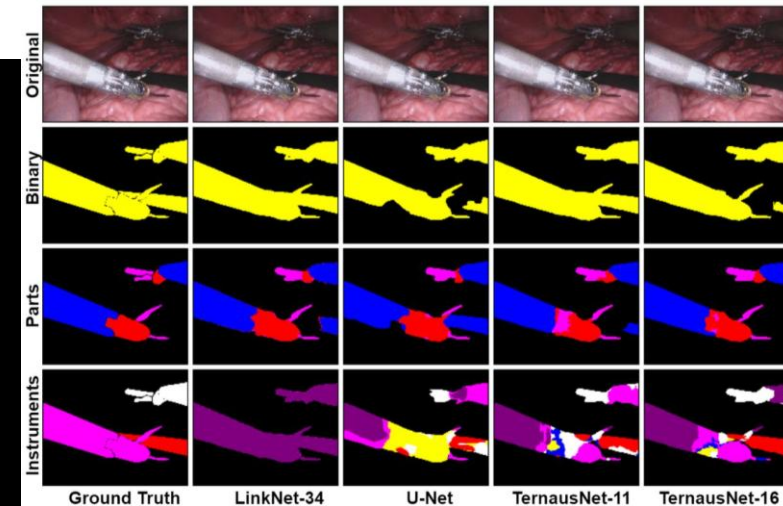
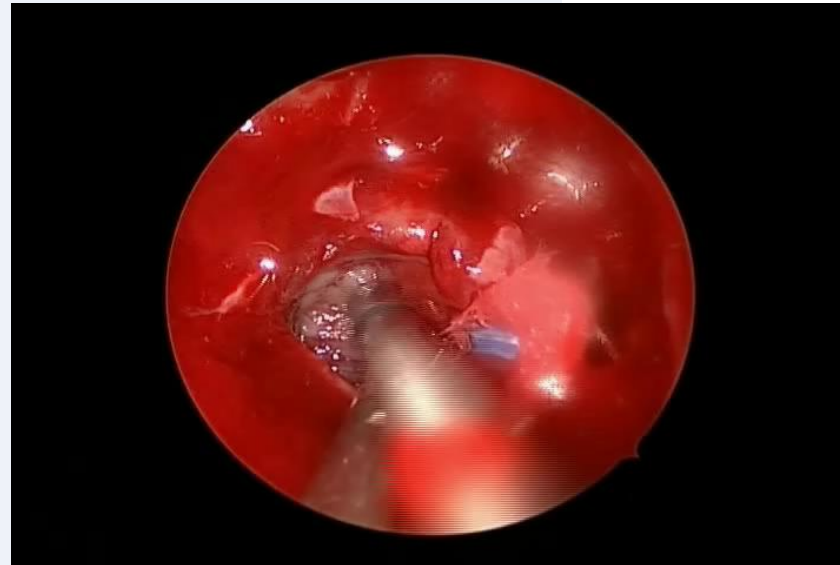
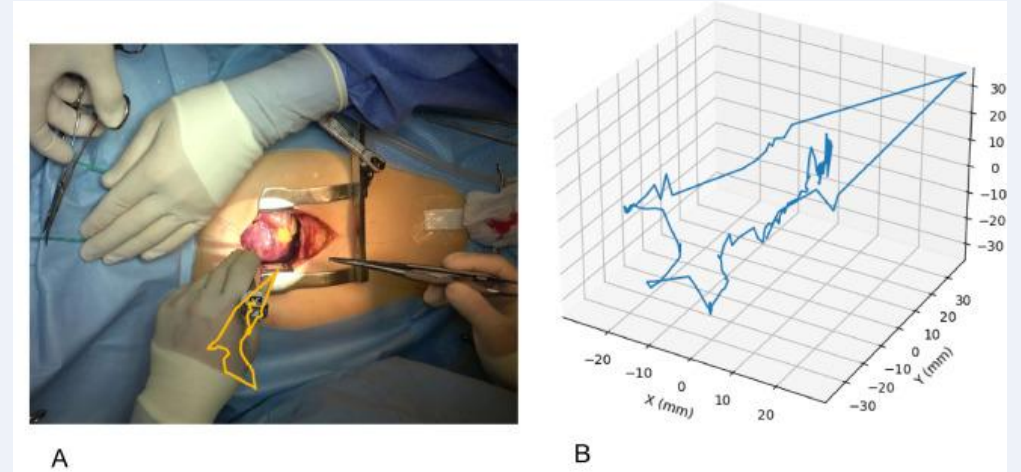
Performance Assessment



Performance Assessment

Tool tracking, motion tracking

- Task completion time
- Motion smoothness
- Position/angles
- Force
- Velocity
- Acceleration

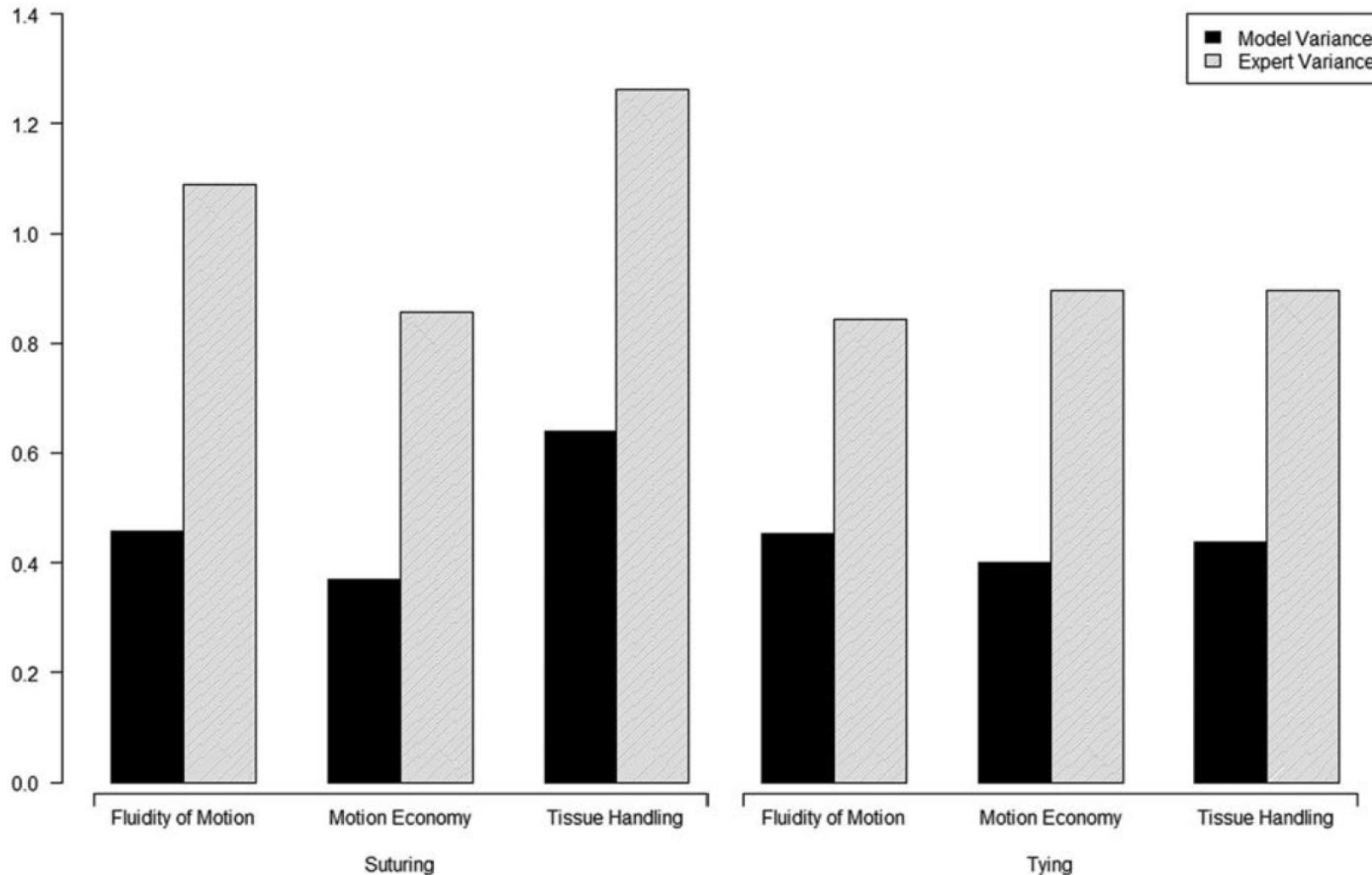


Tool Tracking, Motion Tracking



Performance Assessment

Comparison of Variance Between Experts and Video Based Model



Model developed using kinematic variables. E.g.:

- Speed
- Acceleration
- Idle time
- Working area
- Path curvature

OSATS - Expert rated:

- Fluidity of Motion
- Motion Economy
- Tissue Handling

Performance Assessment

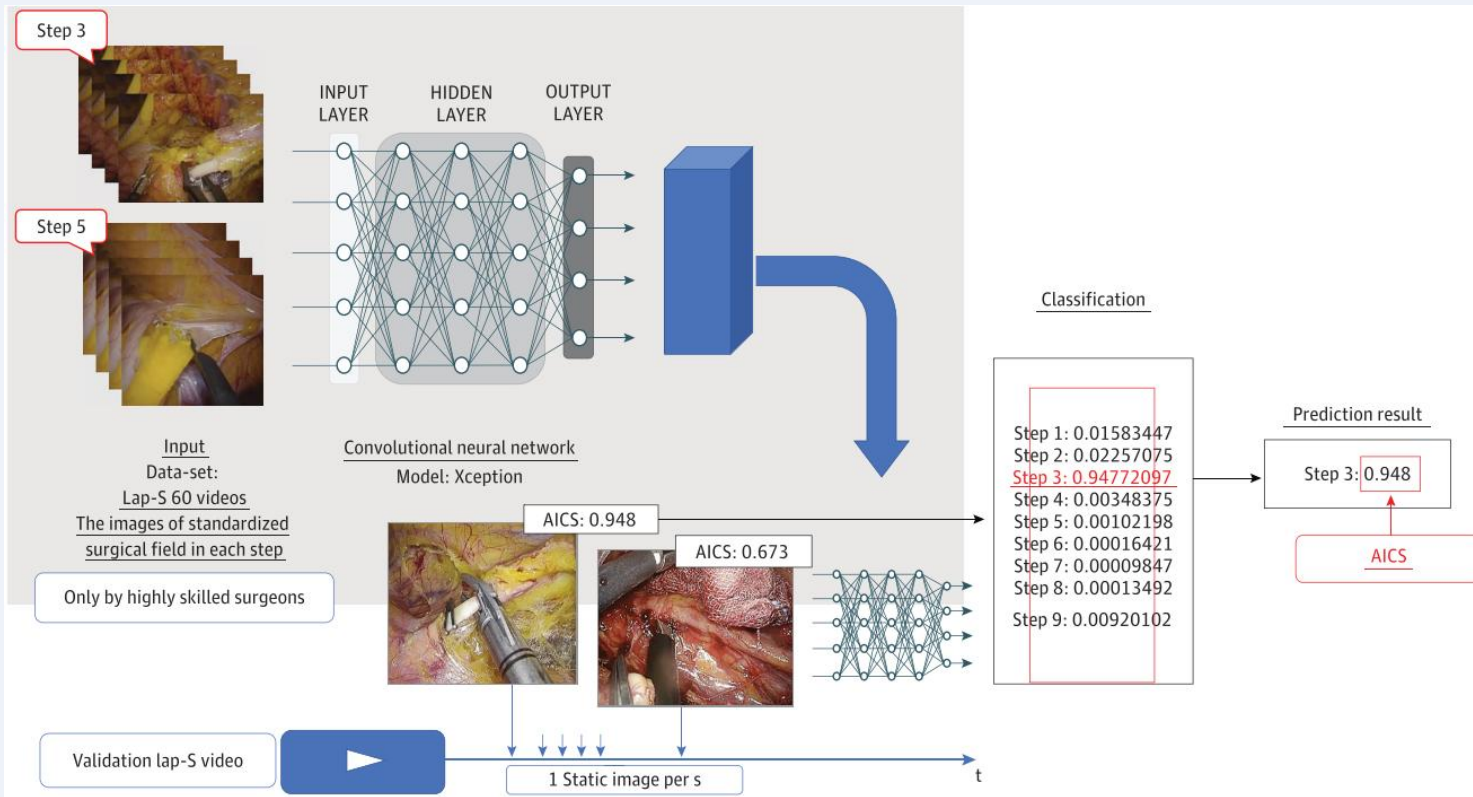
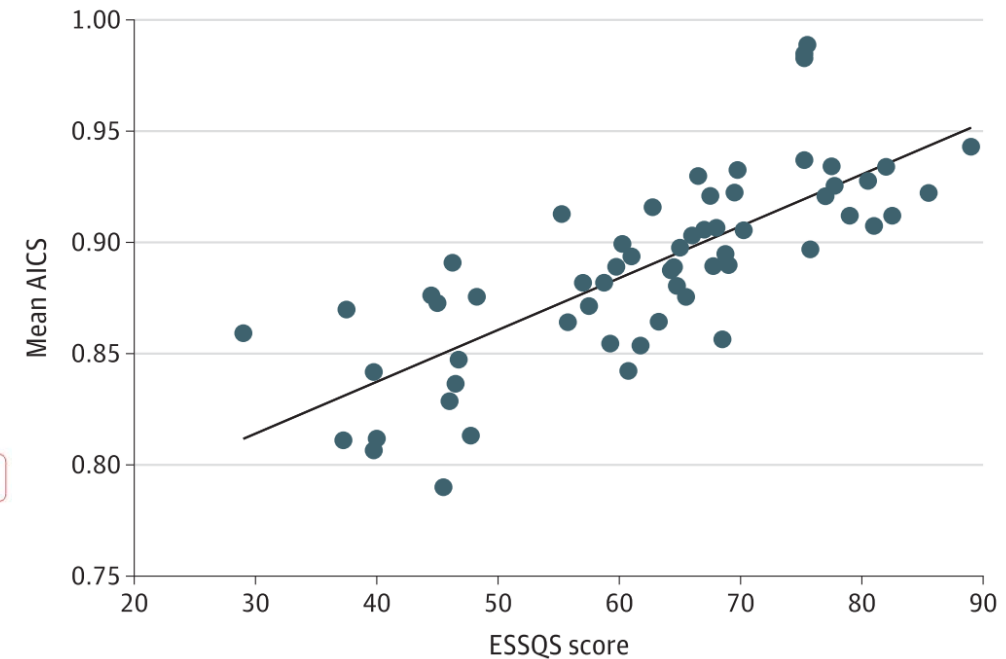
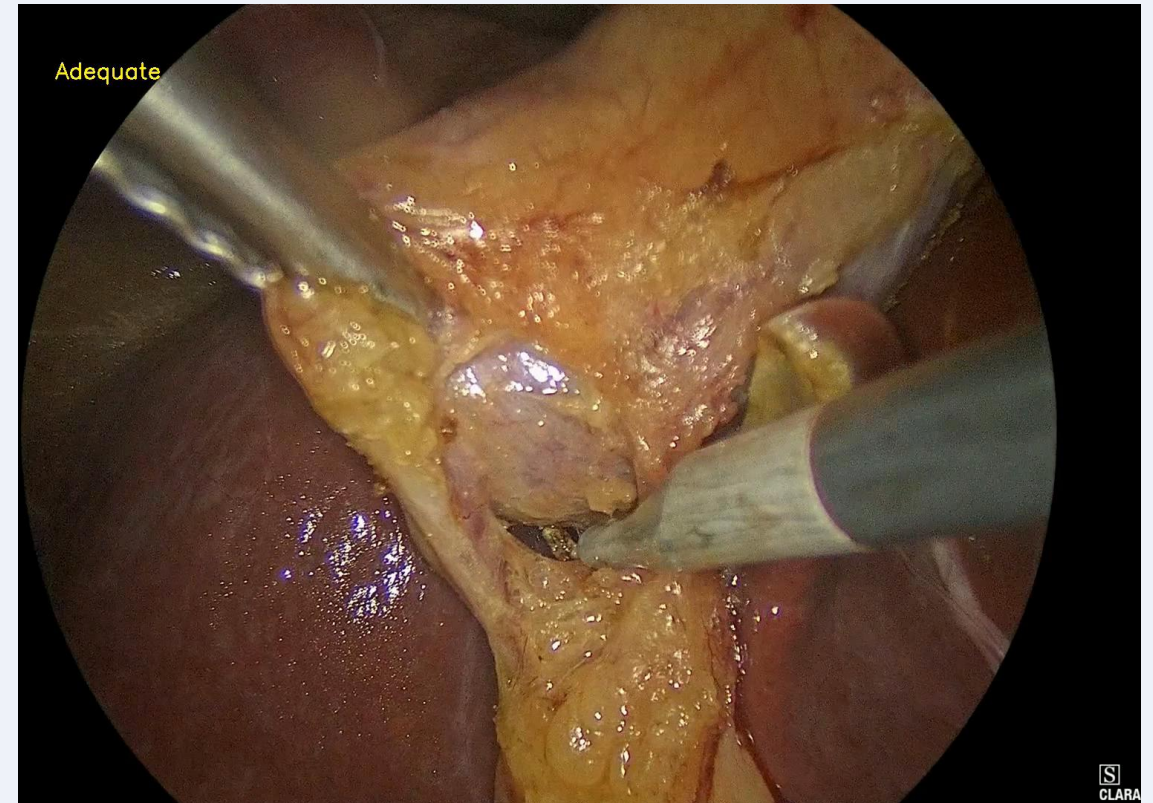
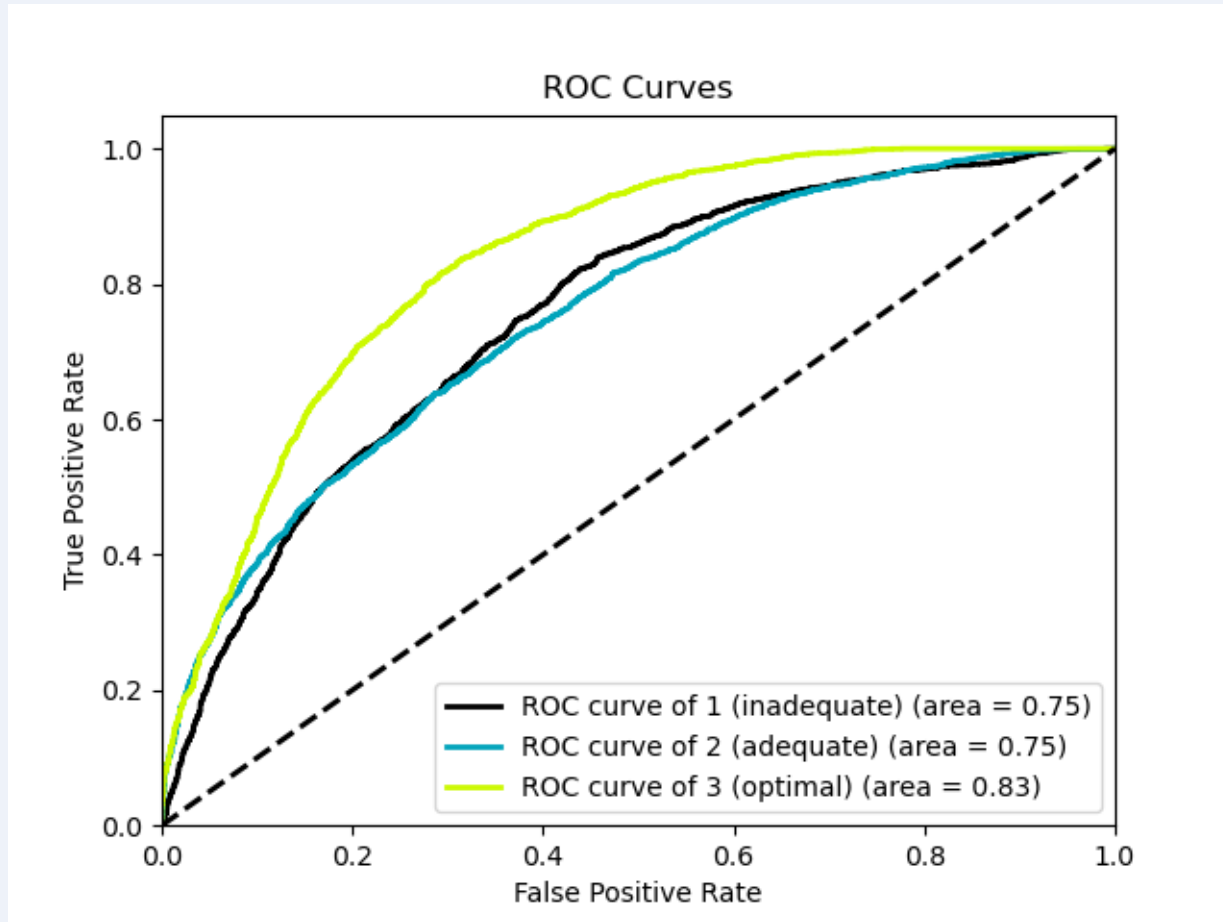


Figure 3. Correlation Between the Artificial Intelligence Confidence Score (AICS) and Endoscopic Surgical Skill Qualification System (ESSQS) Score



AUROC for the low- and high skill group screening were 0.93 and 0.94.

Feedback for Technical Skills – Lap Chole

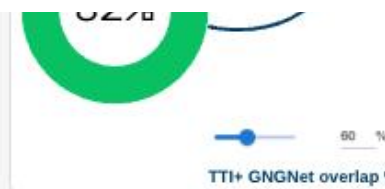
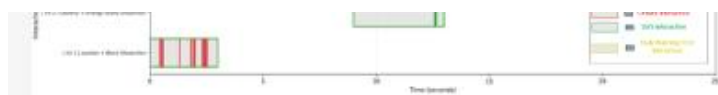
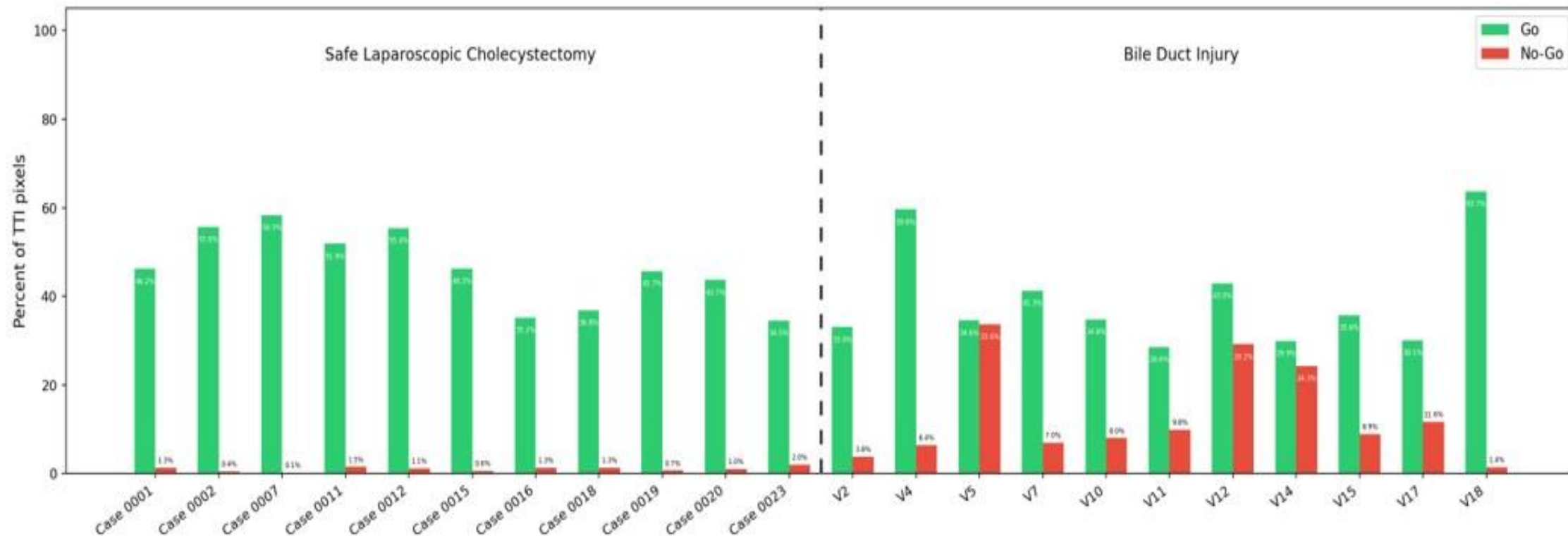


SURGEON TOOL-TISSUE INTERACTION SCORE CARD

TTIs enter the data era...

Frame-level
Visual Feature

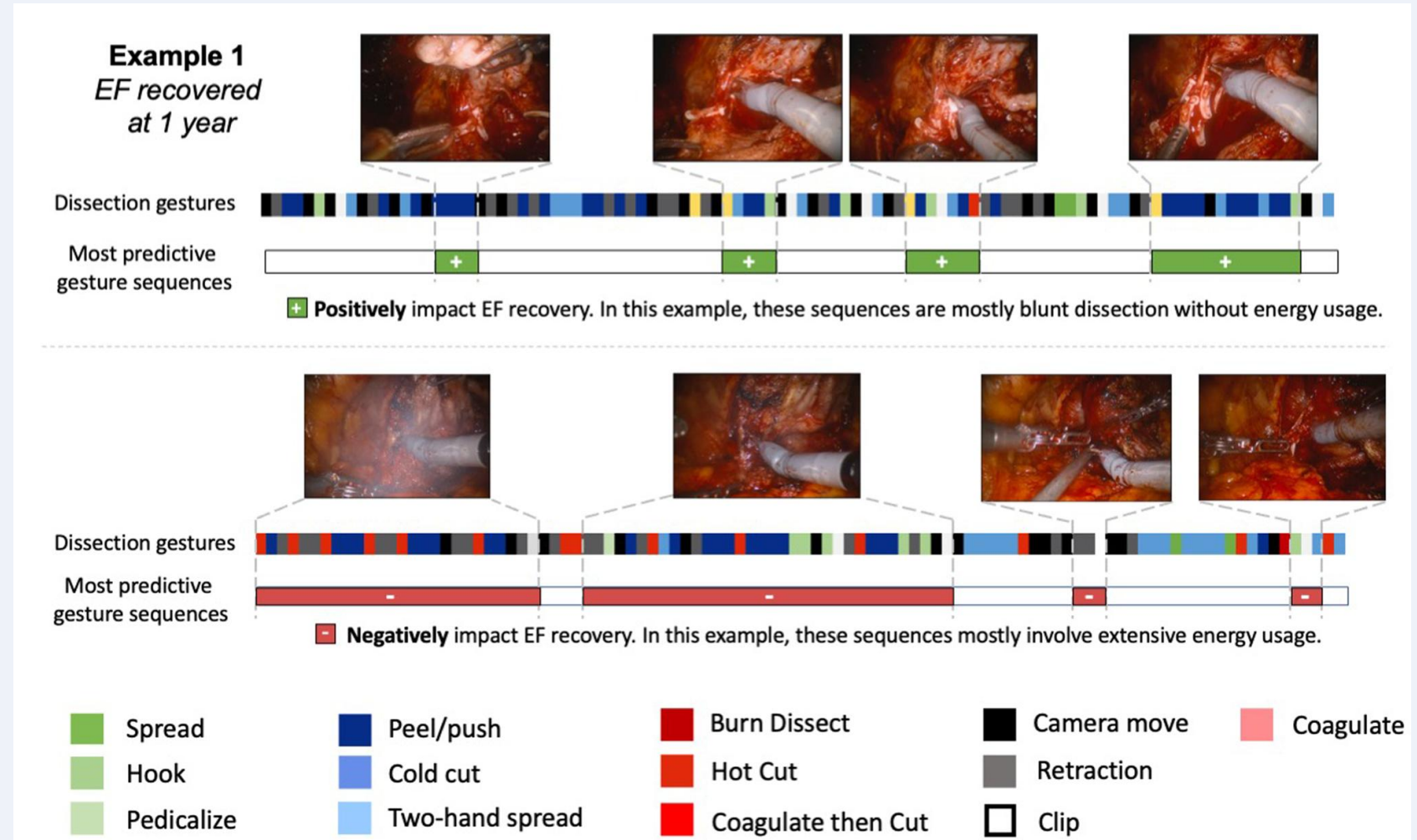
Mean Go / No-Go %



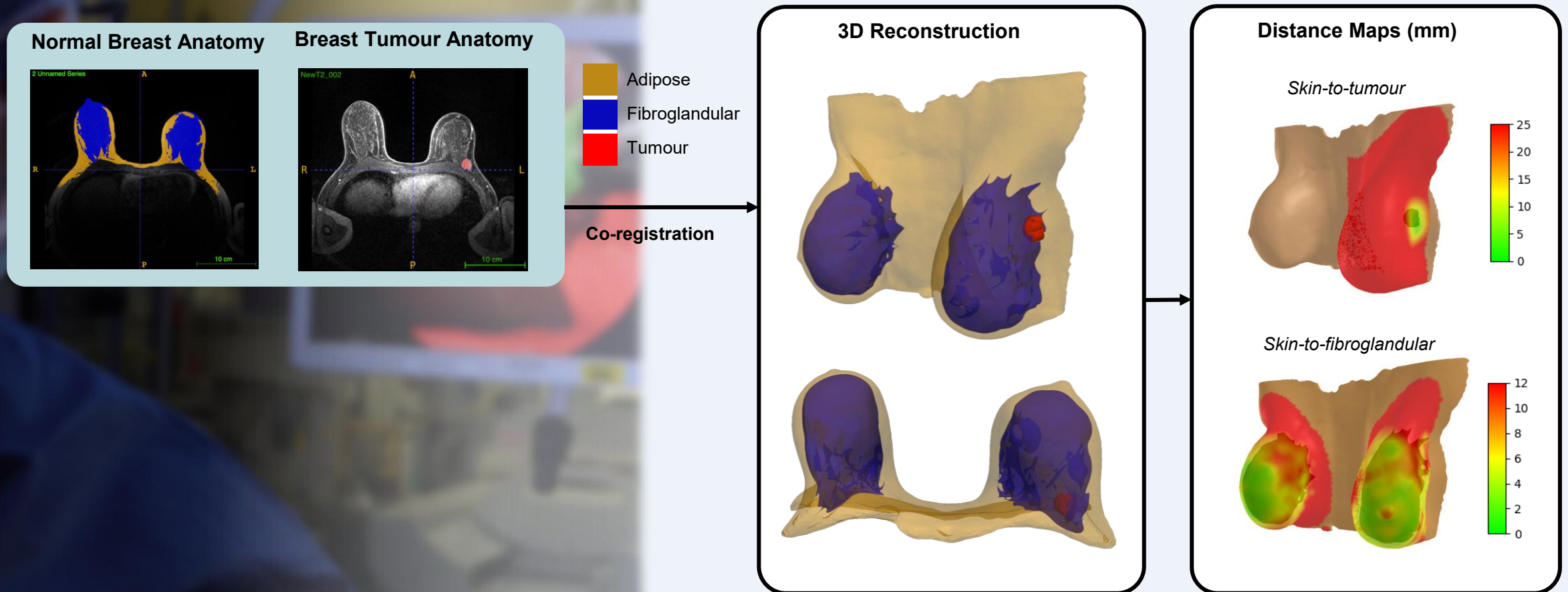
1 Site, Tool Type, Interaction
with Yes/No Temporal
Confirmation

Performance Assessment

- 80 nerve-sparing radical prostatectomies
- 34,323 gestures
- 1-year erectile function recovery



Use-Case 2: Digital Twins for Preoperative Planning and Simulation



Challenges for 3D Rendering

- Automated annotation of anatomical structures
- Semi-automated annotation of nuanced anatomical structures
- Integrated with PACS
- ML and 3D rendering pipeline
- Deployment infrastructure

Infrastructure and PACS Integration

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dev

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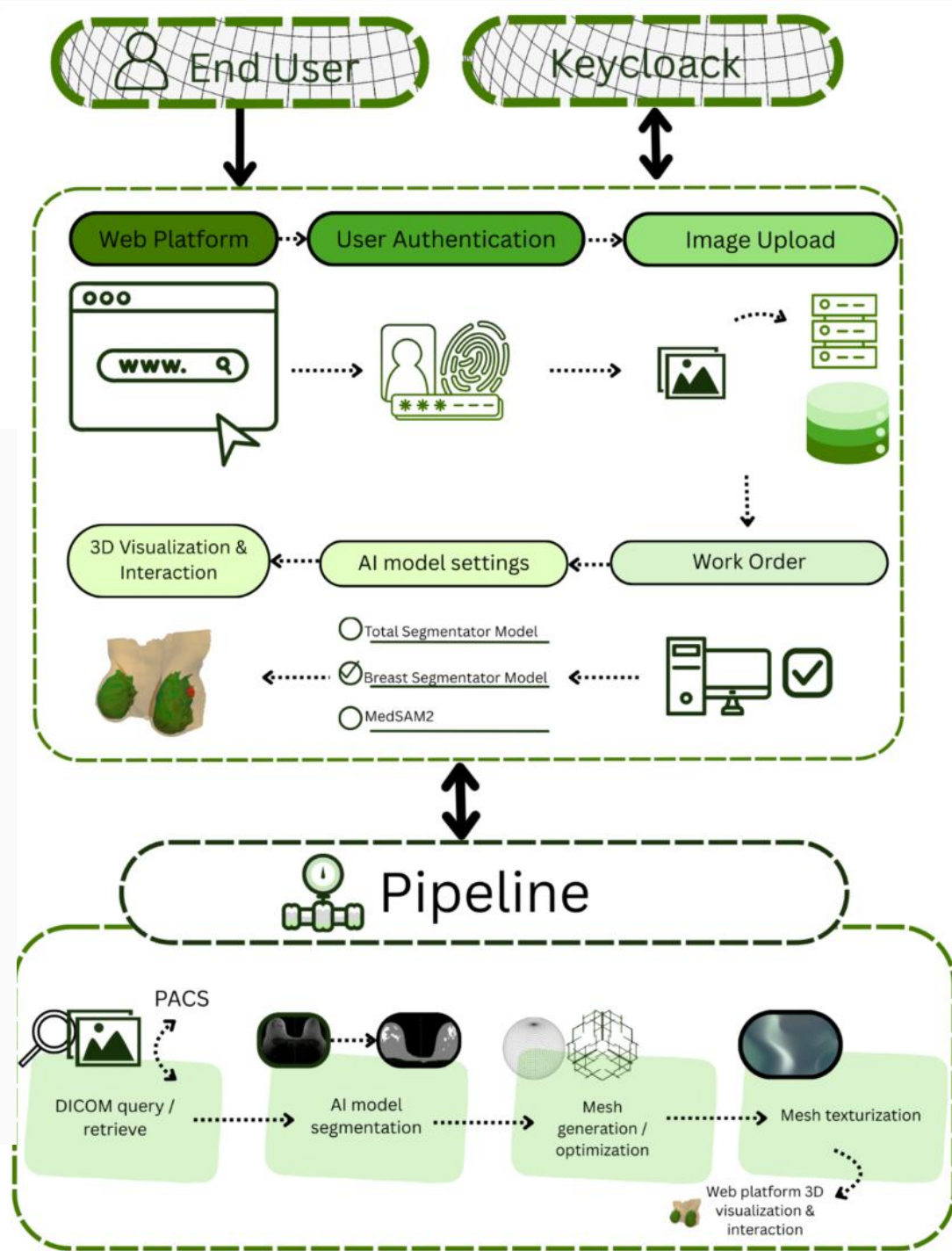
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Work Orders Dashboard

New Work Order

View and manage your surgical planning work orders

Your Work Orders

Work Order ID	Status	Created	Last Updated	Actions
WO-61f8c33b	COMPLETED	1/12/2026, 10:47:13 AM	1/12/2026, 10:48:42 AM	<button>View Results</button>
WO-41837a7b	COMPLETED	1/12/2026, 10:10:57 AM	1/12/2026, 10:13:29 AM	<button>View Results</button>
WO-82f28fe3	COMPLETED	1/12/2026, 9:46:44 AM	1/12/2026, 9:48:01 AM	<button>View Results</button>
WO-326d1681	COMPLETED	1/12/2026, 9:38:59 AM	1/12/2026, 9:40:24 AM	<button>View Results</button>
WO-36bc168c	COMPLETED	1/12/2026, 1:54:03 AM	1/12/2026, 1:55:24 AM	<button>View Results</button>
WO-b91b6748	COMPLETED	1/12/2026, 1:46:53 AM	1/12/2026, 1:48:16 AM	<button>View Results</button>
WO-2f66218b	COMPLETED	1/12/2026, 1:41:52 AM	1/12/2026, 1:43:17 AM	<button>View Results</button>
WO-e51453d9	COMPLETED	1/12/2026, 1:31:13 AM	1/12/2026, 1:33:24 AM	<button>View Results</button>
WO-441e8f10	COMPLETED	1/12/2026, 1:09:53 AM	1/12/2026, 1:11:03 AM	<button>View Results</button>
WO-1992bd71	COMPLETED	1/12/2026, 12:55:56 AM	1/12/2026, 12:57:19 AM	<button>View Results</button>
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WO-12491e2f	COMPLETED	1/11/2026, 11:57:27 PM	1/12/2026, 12:01:18 AM	<button>View Results</button>
WO-9121a275	COMPLETED	1/11/2026, 11:53:10 PM	1/11/2026, 11:53:46 PM	<button>View Results</button>

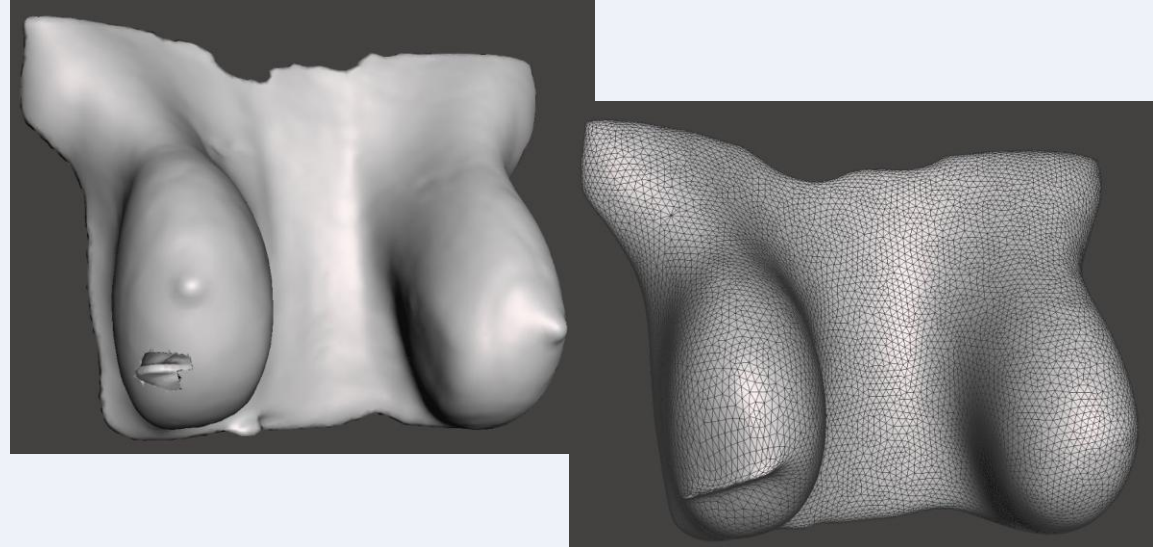
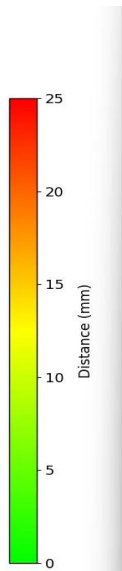

SURGICAL ARTIFICIAL INTELLIGENCE
RESEARCH ACADemy

Username

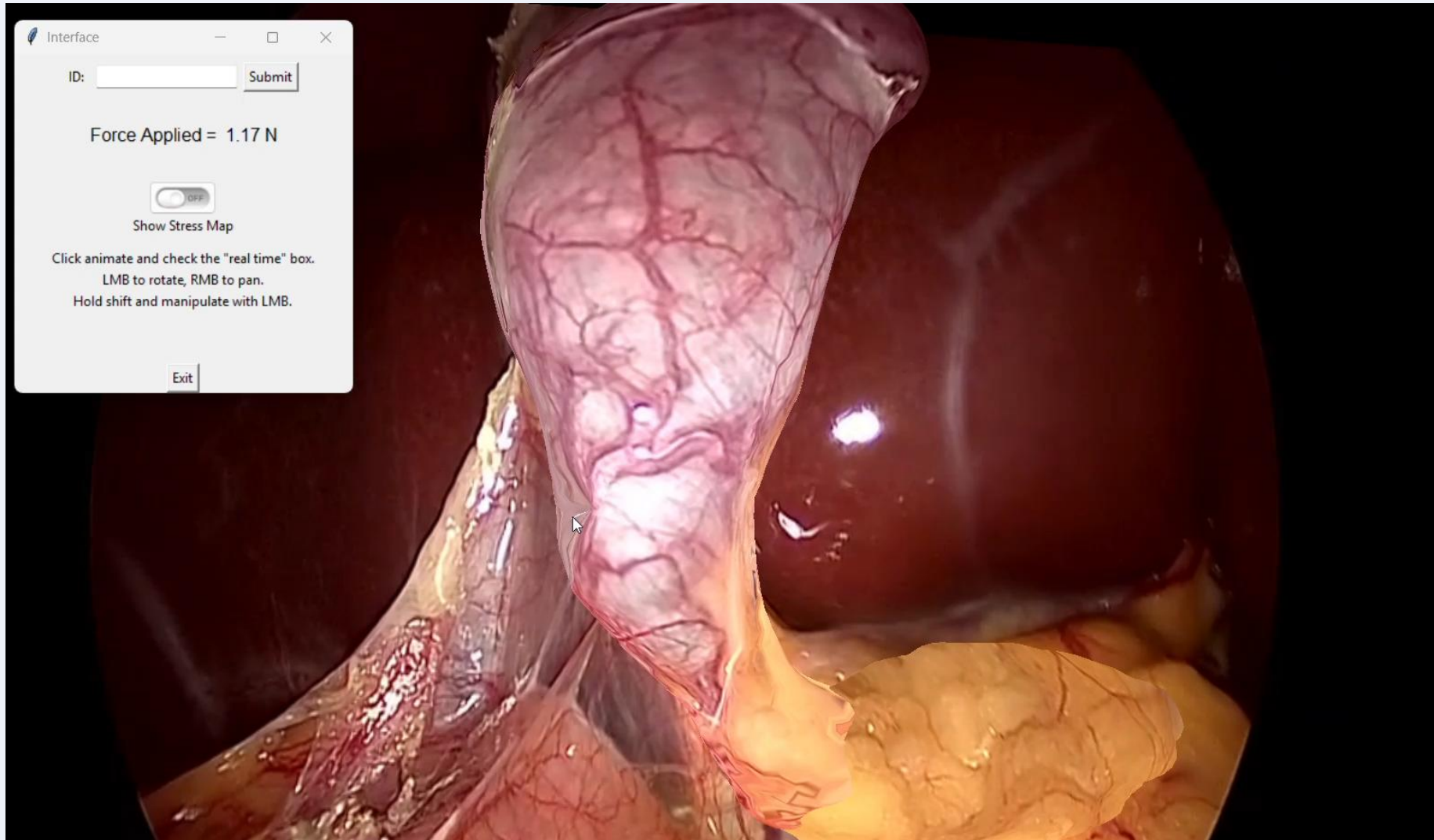
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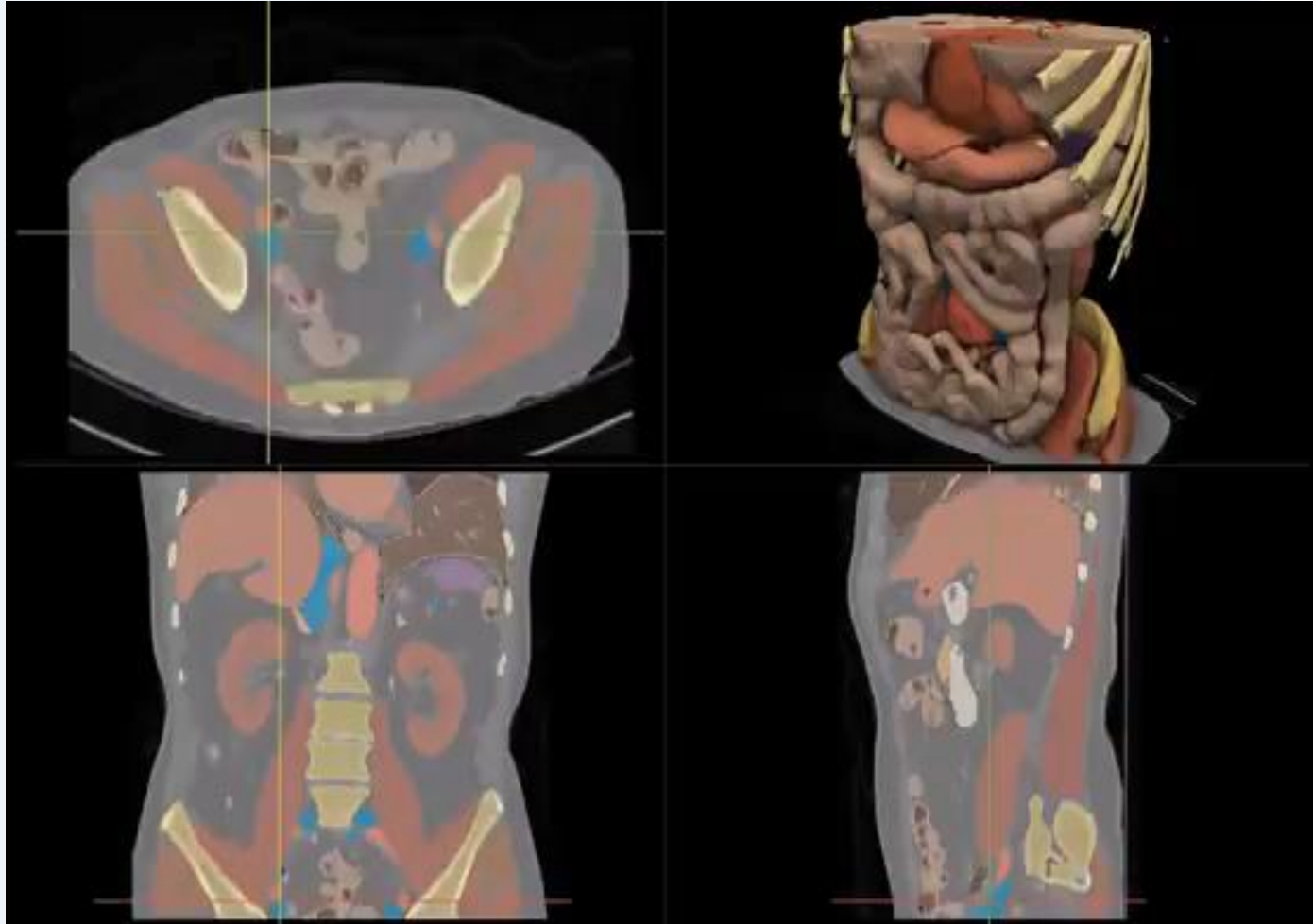
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Simulation – 3D Modelling



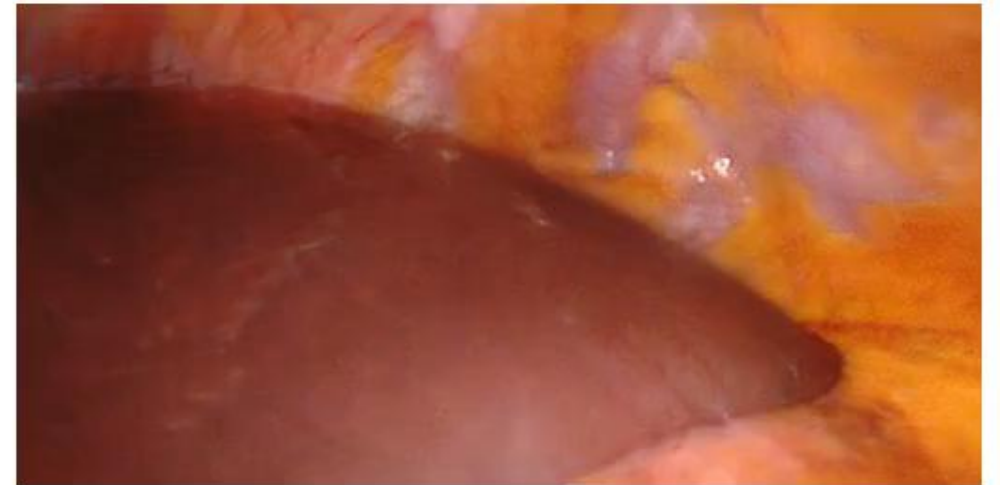
Simulation – Synthetic Data



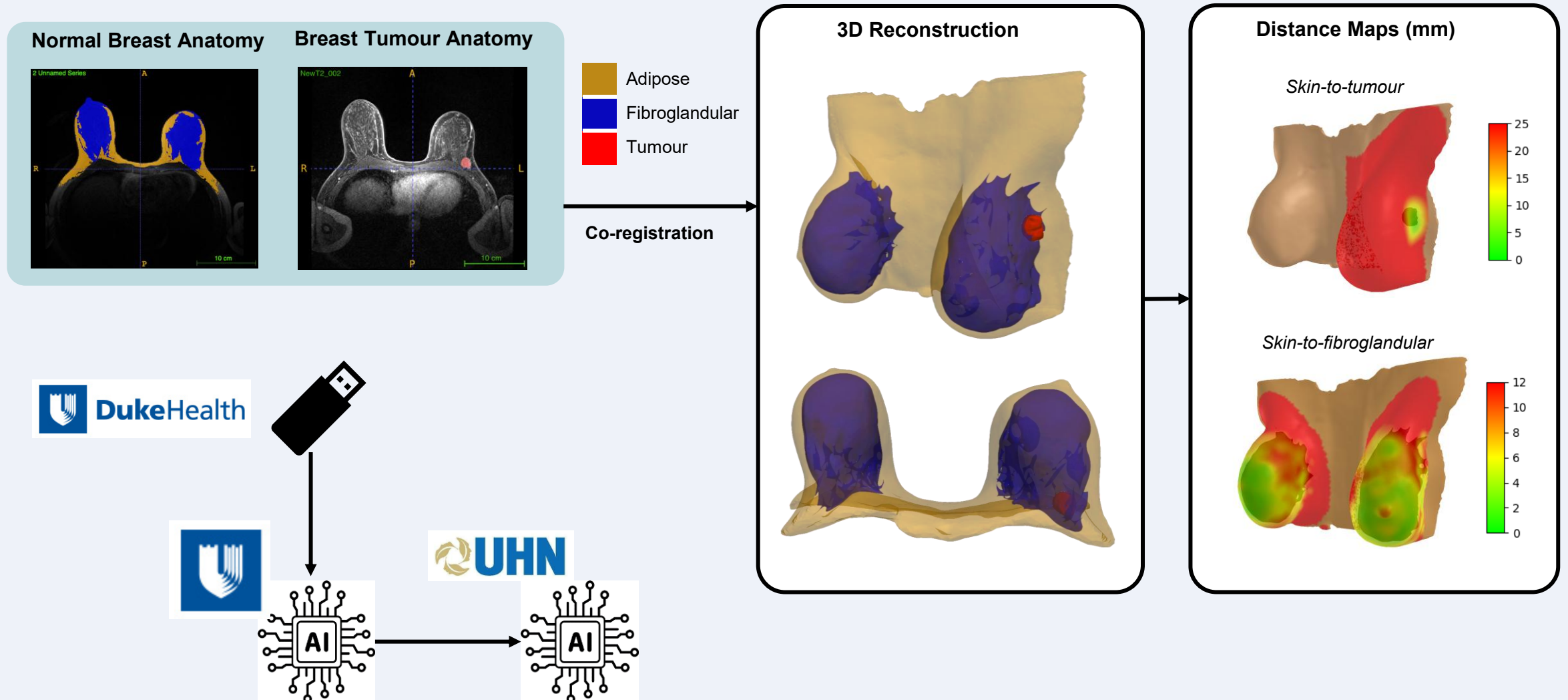
Input



Output



Generalizability of Surgical AI

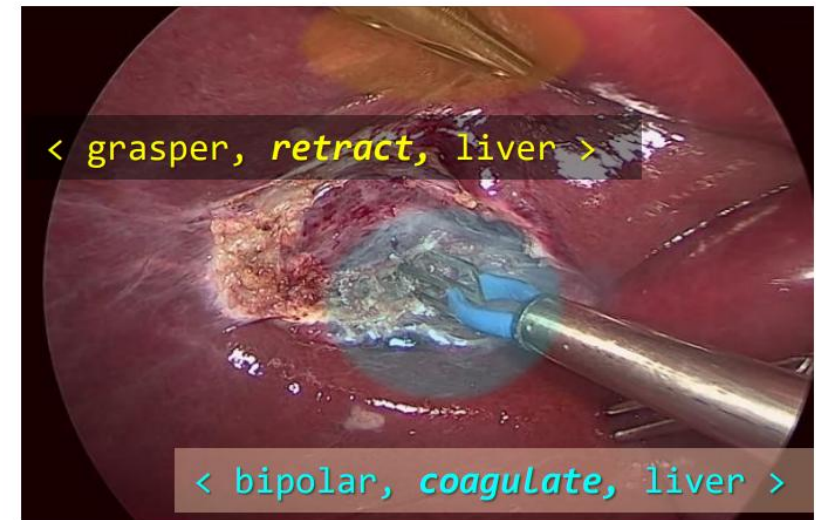
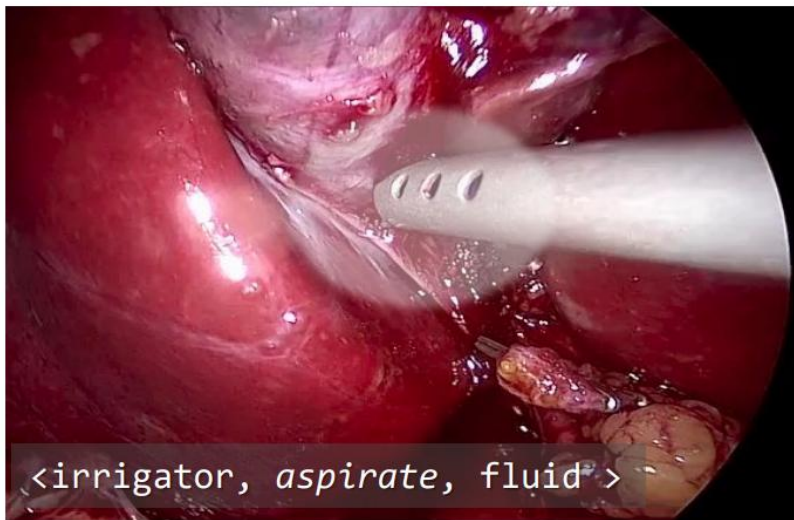
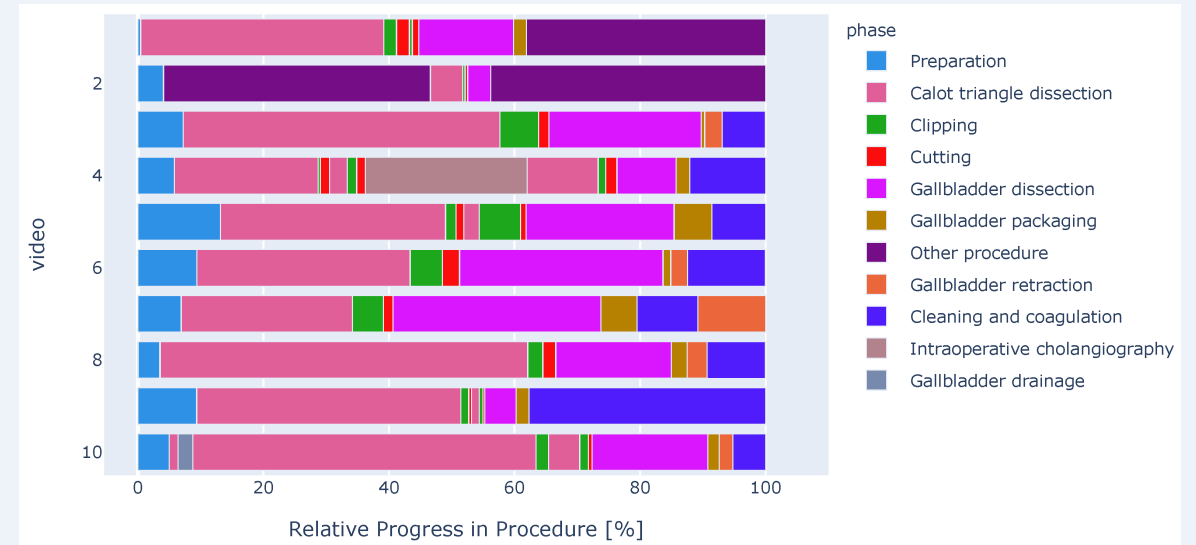


Use-Case 3: AI for Workflow and Crisis Management



Surgical AI for OR Workflow

- Surgical phase recognition
- MM-OR: multimodal holistic operating room modelling
- “Action-Triplet” algorithms:
 - Instrument – Task – Target



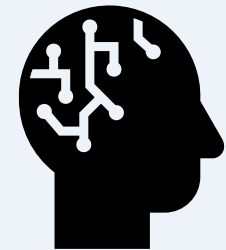


Trials
Se

Patient
○ Nu
○ M
○ Se

MasTER

- Multiplayer, web-based disaster management platform
- Allows for:
 1. High-fidelity simulation training
 2. Real-time operational decision support during live incidents/MCI/public health emergencies
- Deployment across civilian public health systems and military operational environments
- Incorporates AI to optimize workflow and decision support



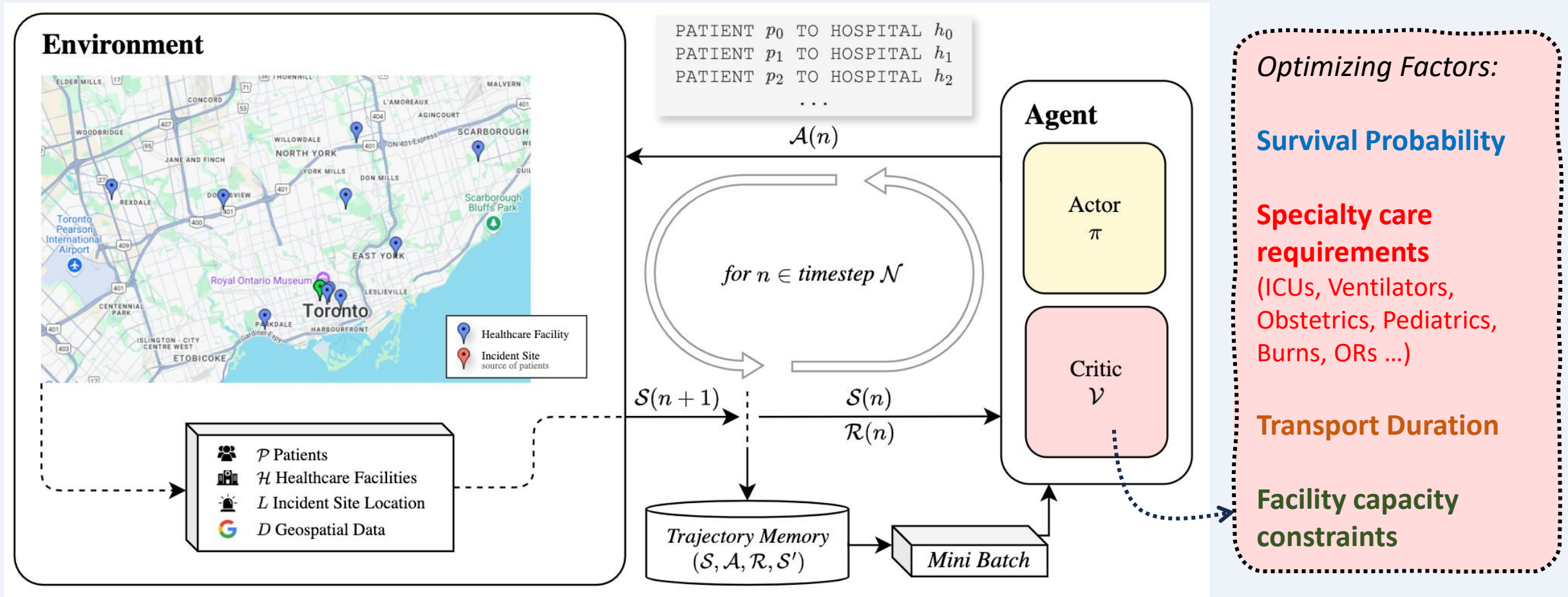
MasTER – Core Capabilities

AI-Augmented Triage & Transfer Optimization

Embedded algorithms provide decision support by:

- Matching injury severity and mechanism to appropriate facility capabilities
- Optimizing travel time and resource allocation
- Directing critically unstable patients to Level I trauma centres
- Routing specialty injuries (e.g., burns, neurotrauma) to appropriate centers
- Dynamically adjusting recommendations as resource availability changes

Surgical AI for Patient Triage



The AI enhances — not replaces — clinical judgment, accelerating high-stakes decisions in time-sensitive environments



Toronto Pearson International Airport
Toronto, Canada

Notifications



1/6/2026, 8:44:32 PM

Session started: A MCI incident with 20 patients has been initiated in Toronto.

Untriaged Patient



Name: Durand Jones
Age: 12
Gender: Male
Pregnancy: No
Description: Brown hair, black eyes, height 151cm; broken right arm.

← Previous

Patient 1 of 5

Next →



AI Assistant

Hello! I am SARA MaSTER, your AI assistant for this simulation.

Invalid Date

Should this patient go to a level 1 facility?

1/6/2026, 8:45:06 PM

No. A 12-year-old with a single broken right arm does not meet the criteria for a Level 1 trauma center. He can be treated at a local hospital that has pediatric orthopedic services. Transfer to a Level 1 facility is only warranted if additional injuries, significant pain, or complications (e.g., neurovascular compromise, open fracture, or need for advanced imaging/surgery) are identified.

1/6/2026, 8:45:09 PM

Ask about anything that's in your mind.

Agent Mode

AI Triage



Walkie-Talkie

0/3

0/5

0/11

1

00:00:47

Patient Data Editor

Confirm Triage

Required Information

Patient ID *

Name *

Age *

Gender

Severity Level *

Patient Location *



Search

Search and select the patient's current location. *



University of Toronto, St. George Campus
Toronto, Canada

Notifications

1/6/2026, 8:54:17 PM
Session started: A MCI incident with 50 patients has been initiated in Toronto.

0/100/200/2000

Anna Petrova

ID: OBA0E632

CriticalAge: 11Female00:05:05

Patient with Critical injuries from MCI

Required Facility Capabilities

Burn	✓	Trauma Center	✓	Emergency	✓
Obstetric	×	Ent	×	Vascular	×
Pediatric	✓	Hepatobiliary	✓	Orthopedic	×
Neurosurgical	×	Ophthalmology	✓	Thoracic	×
Cardiac	✓				

Required Medical Resources

Ordinary ICU	1	Operating Room	0	Ventilator	1
Interventional Radiology	0	Oxygen Cylinder	0	PRBC Unit	0
Isolation	0	Ward	1	Decontamination Unit	0

Find address or place

Unassigned Patients

Search patients, injuries, resources...

Sort BySeverityTime to Death

Anna Petrova

ID: OBA0E632

Critical

00:05:05

Age: 11Female

David Johnson

ID: FEAD433F

Critical

00:10:45

Age: 33Male

Anna Ivanov

ID: D9EDBEFC

Critical

00:11:12

Age: 60Female

Andriy Kovalenko

ID: OC20D8C6

Critical

00:12:25

Age: 50Male

Maria Ivanov

ID: D9EDBEFC

Critical

Healthcare Facilities

Search facilities, resources...

Sort ByLevelETA

Sunnybrook Health Sciences Centre

Level 1

Patients: 0Trauma Center

The Hospital for Sick Children

Level 1

Patients: 0Trauma Center

St. Michael's Hospital

Level 1

Patients: 0Trauma Center

Toronto Western Hospital

Level 2

Patients: 0Trauma Center

erson International Airport
da

I incident with 20 patients has to.

Untriaged Patient

Name: Durand Jones

Age: 12

Gender: Male

Pregnancy: No

Description: Brown hair, black eyes, height 151cm; broken right arm.

← Previous

Patient 1 of 5

Next →

AI Assistant

Ask about anything that's in your mind.

Agent Mode

AI Triage

Walkie-Talkie

Patient Data Editor

Confirm Triage

Required Information

Patient ID *Name *

Age *Gender

Severity Level *

Patient Location *

Patients Overview

Sofia Kuznetsova

ID: 59D2141E

Severe

Age: 63

Female

00:08:18

Patient with Severe injuries from MCI

Location

Required Facility Capabilities

Burn	✓	General Surgery	×
Orthopedic	×	Trauma Center	×
Neurosurgical	×	Ophthalmology	×
Obstetric	×	Ent	×
Thoracic	×	Vascular	×
Pediatric	×	Cardiac	×

Required Medical Resources

Ordinary ICU	0	Operating Room	0
Ventilator	0	Interventional Radiology	0
Oxygen Cylinder	1	PRBC Unit	2
Isolation	0	Ward	0

Unassigned

Natalia Shevchenko

ID: 1DA882F

Critical

Age: 90Female

Alex Brown

ID: C934AF17

Critical

Age: 44Male

James Kovalenko

ID: 2A009C30

Critical

Age: 50Male

Michael Johnson

ID: 2E34550E

Critical

Age: 81Male

Sofia Ivanov

ID: 601ECAC6

Critical

Age: 51Female

Oleh

ID: 601ECAC6

Critical

Age: 51Female

Awaiting Transfer

In-Transfer

Sofia Kuznetsova

ID: 59D2141E

Severe

Age: 63Female

Arrived

David Johnson

ID: 70F5B8EC

Critical

Age: 16Male

Dmytro Jones

ID: 95275318

Critical

Age: 90Male

Michael Kuznetsova

ID: 728F662B

Critical

Age: 62Male

CN Tower
Toronto, Canada

Notifications

2025-09-16, 7:03:36 p.m.
Patient Victoria Smith (ID: 6767A59F) has deceased due to untimely treatment.

2025-09-16, 6:51:05 p.m.
Patient Kateryna Williams (ID: EBE44B2F) has deceased due to untimely treatment.

2025-09-16, 6:31:38 p.m.
Patient James Petrova (ID: BB549E7F) has deceased due to untimely treatment.

2025-09-16, 6:30:16 p.m.
Patient Ivan Johnson (ID: 7E3DBEDB) has deceased due to untimely treatment.

2025-09-16, 6:25:00 p.m.
Session created by test@uhn.ca

0/00/00/14

Click a patient or facility for more details.

Find address or place

Unassigned Patients

Search patients, injuries, resources...

Sort BySeverityTime to Death

Ivan Johnson

ID: 7E3DBEDB

Deceased

Age: 40Male

James Petrova

ID: BB549E7F

Deceased

Age: 53Male

Viktoria Smith

ID: 6767A59F

Deceased

Age: 23Female

Kateryna Williams

ID: EBE44B2F

Deceased

Age: 35Female

Maria Jones

ID: 6767A59F

Minor

Age: 23Female

Healthcare Facilities

Search facilities, resources...

Sort ByLevelETA

Sunnybrook Health Sciences Centre

Level 1

Patients: 0Trauma Center

St. Michael's Hospital

Level 1

Patients: 0Trauma Center

The Hospital for Sick Children

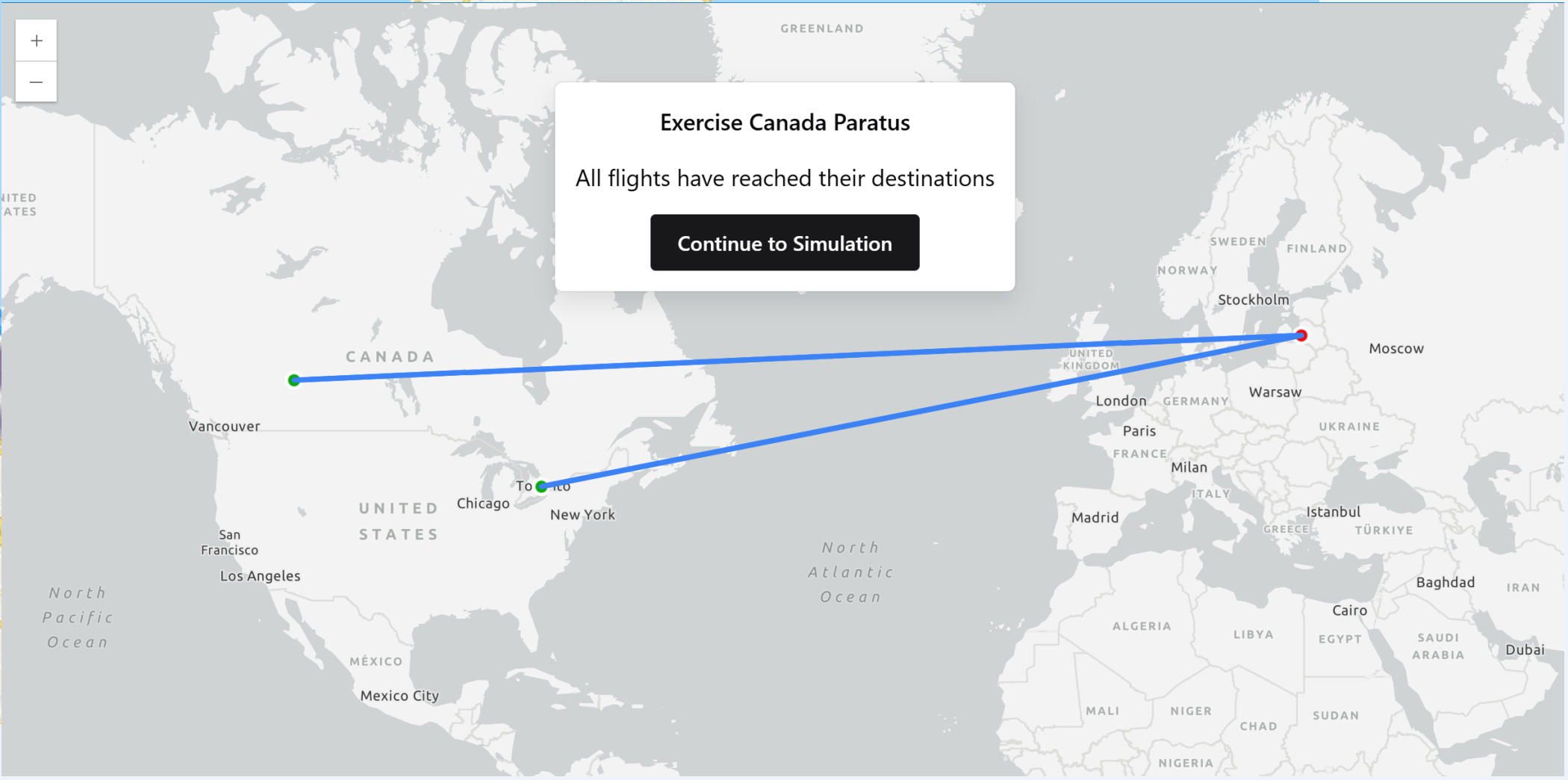
Level 1

Patients: 0Trauma Center

Toronto Western Hospital

Level 2

Patients: 0Trauma Center



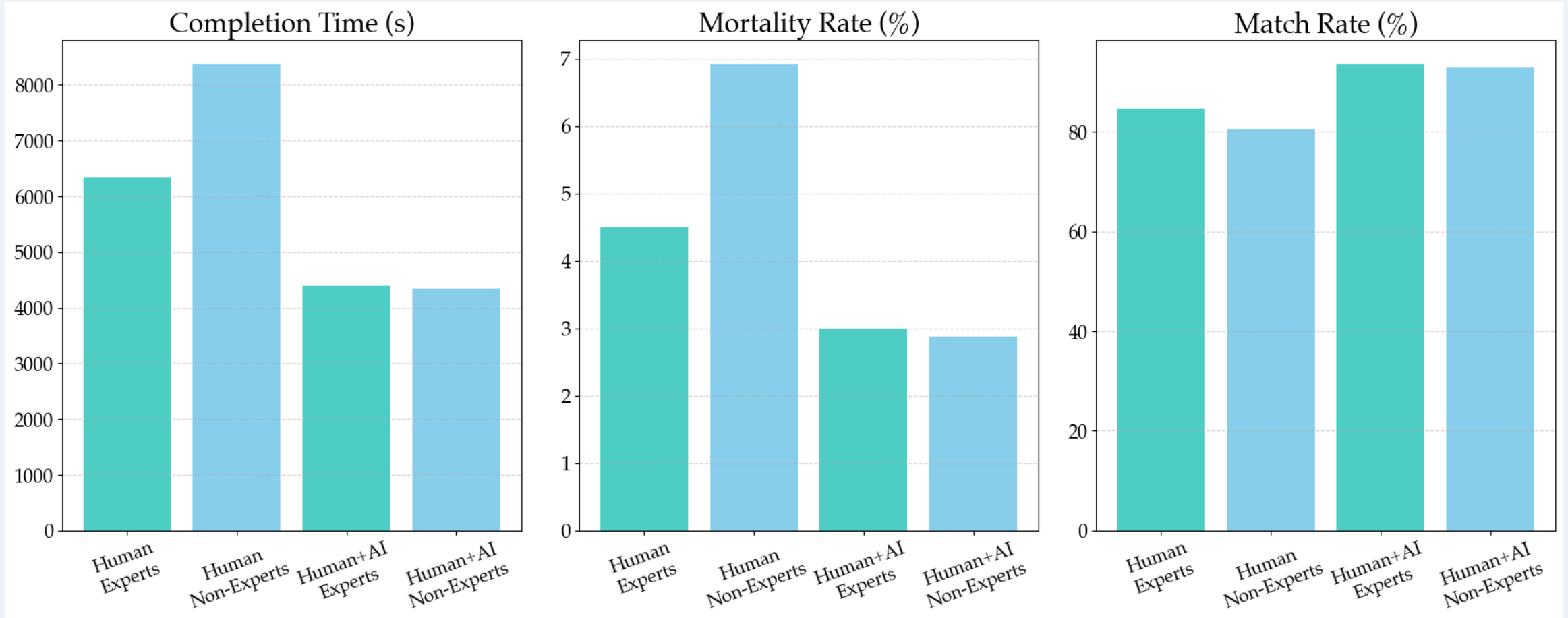
Exercise Canada Paratus

All flights have reached their destinations

Continue to Simulation

Surgical AI for Patient Triage

Experts vs. Non-Experts



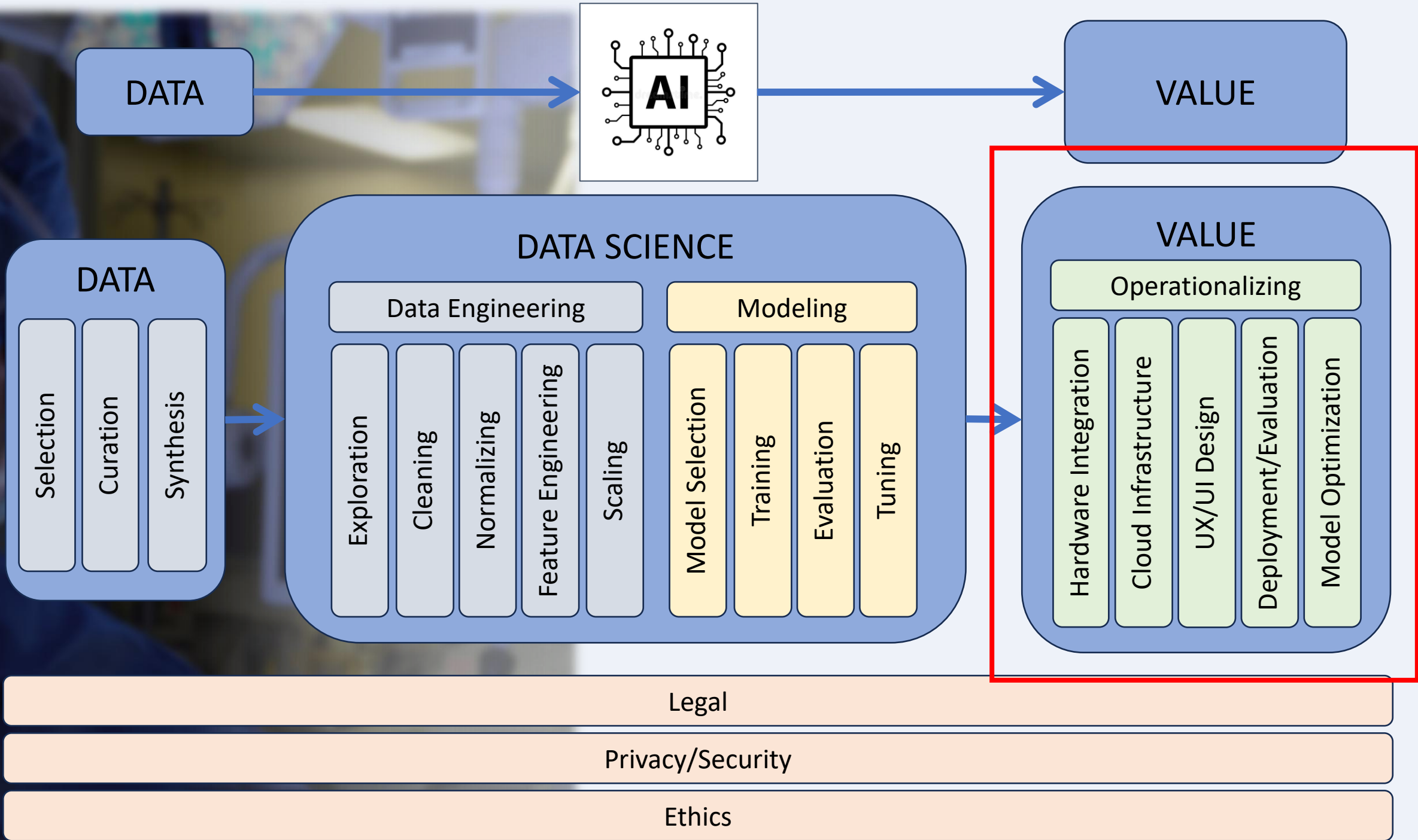
Real-World Uses

- Disaster management planning and simulation for regional healthcare systems
- Military medical personnel training for mass casualty events in combat zones
- Public health emergency planning for pandemics or bioterrorism events (e.g. COVID-19)



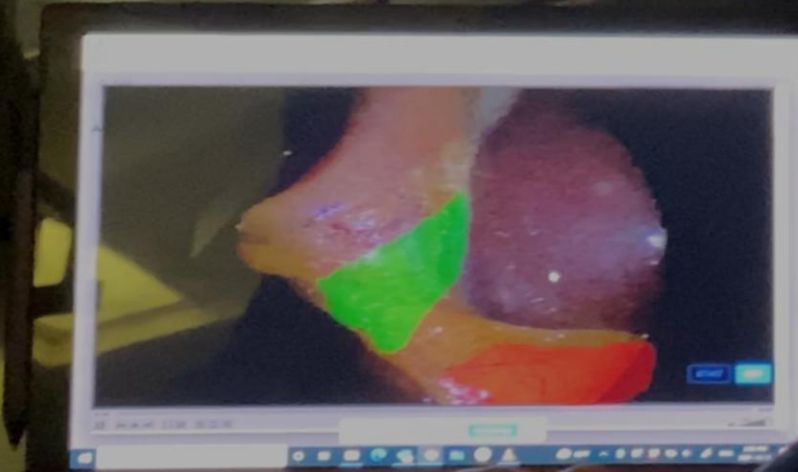


Surgical AI – Hype vs. Reality

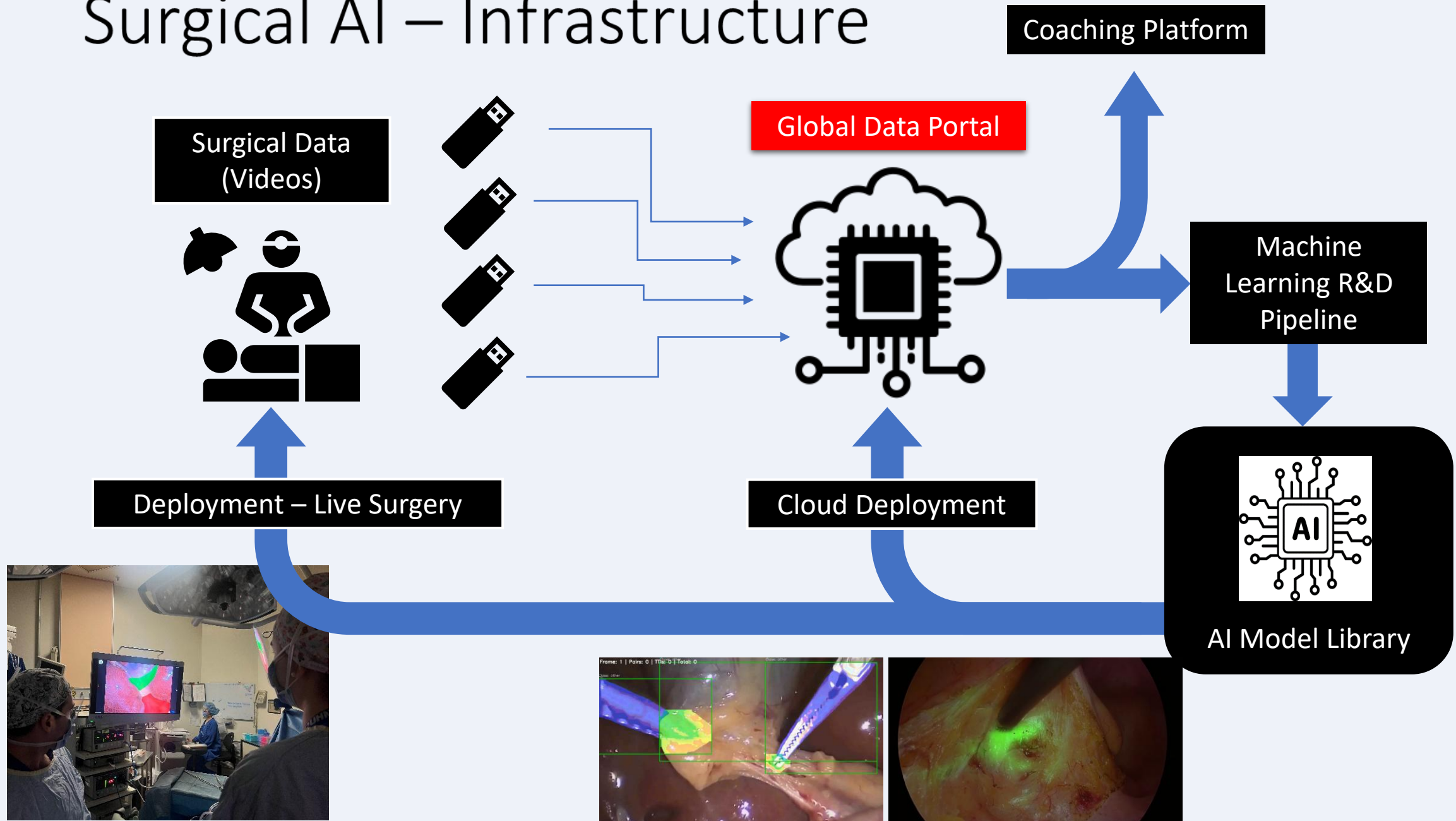


Surgical AI Implementation Challenges

- Infrastructure
- Deployment strategies for real-time support
- Scaling to low resource settings
- Generalizability to local institution
- Model optimization
- Design considerations



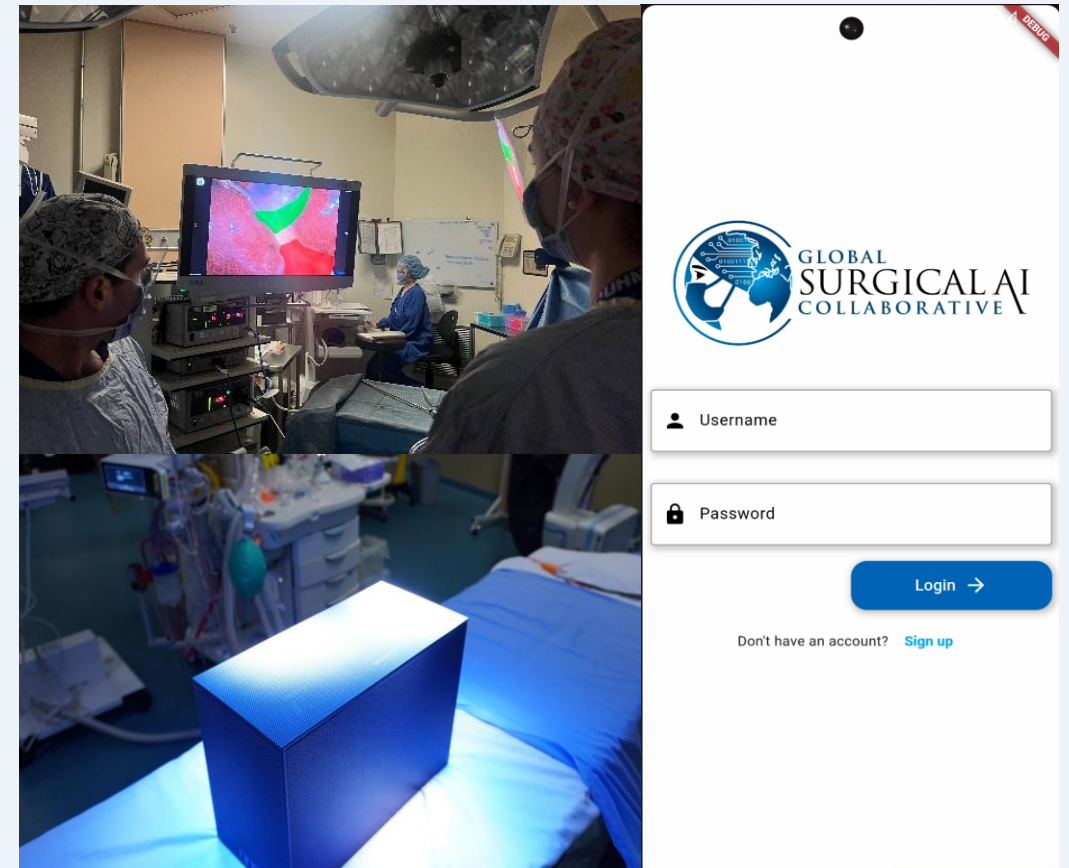
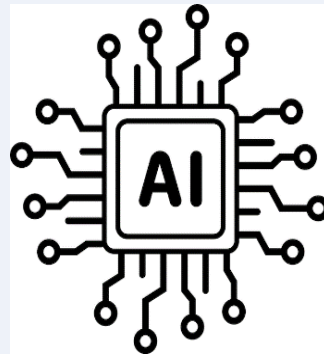
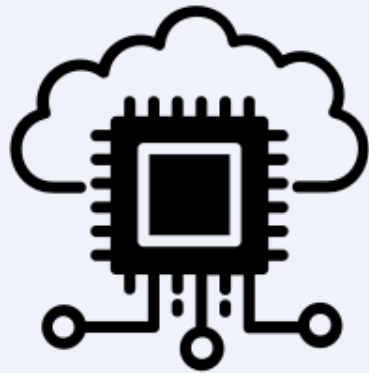
Surgical AI – Infrastructure



Surgical AI – Infrastructure

Surgical AI:

- Raw AI algorithms
- Deployment solution for real-time AI
- Surgical video analysis pipeline

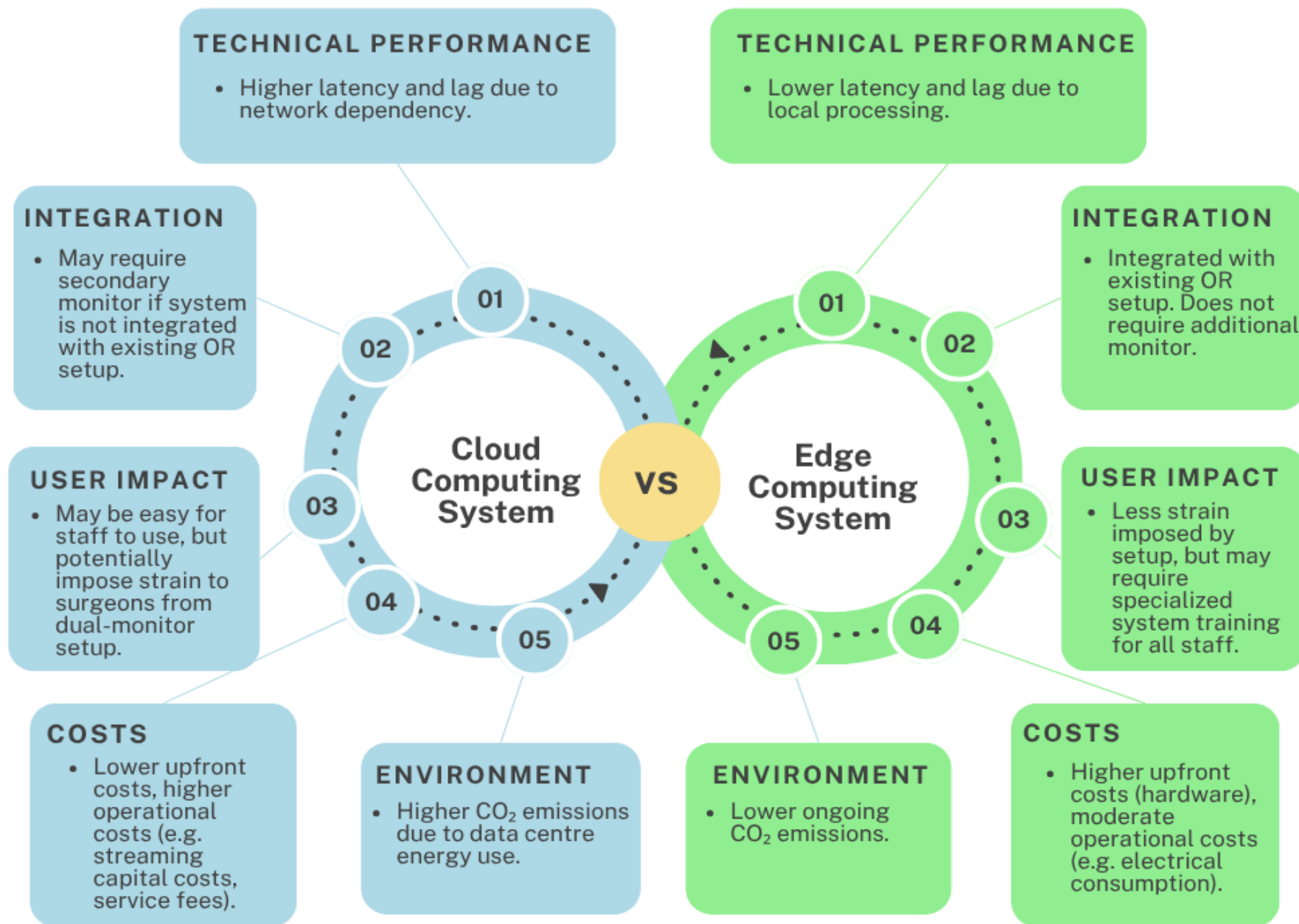


Su

High Ha

Medium

High Maint



Hardware Cost



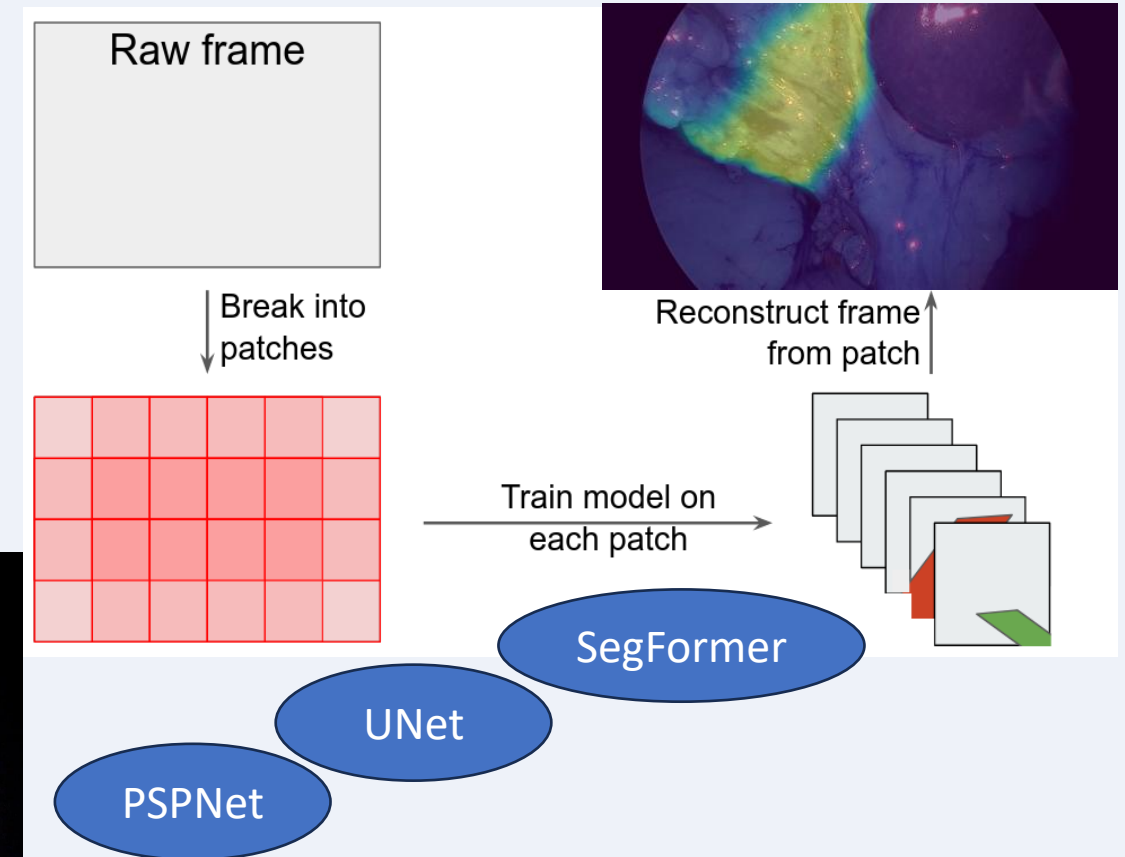
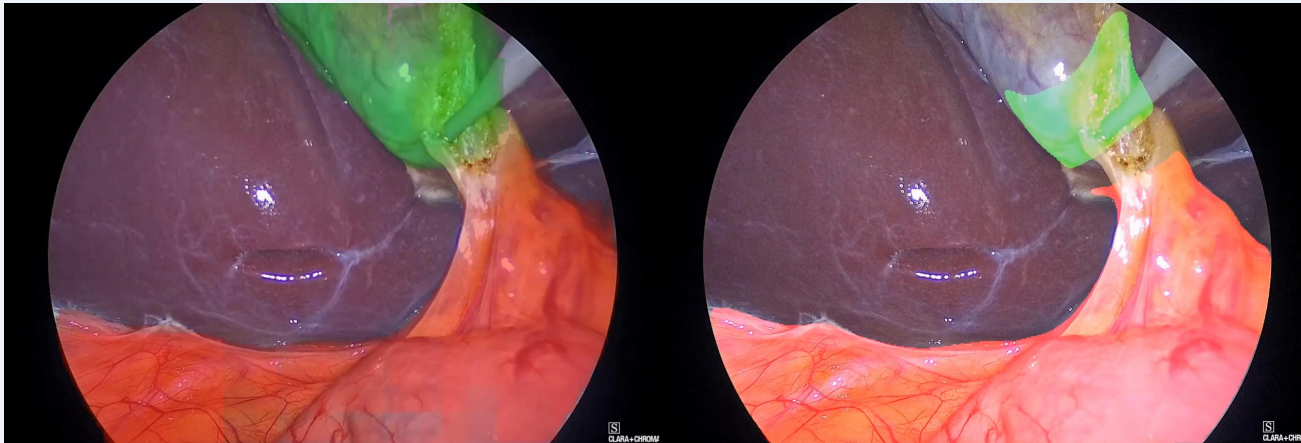
Recovery Cost



Compliance

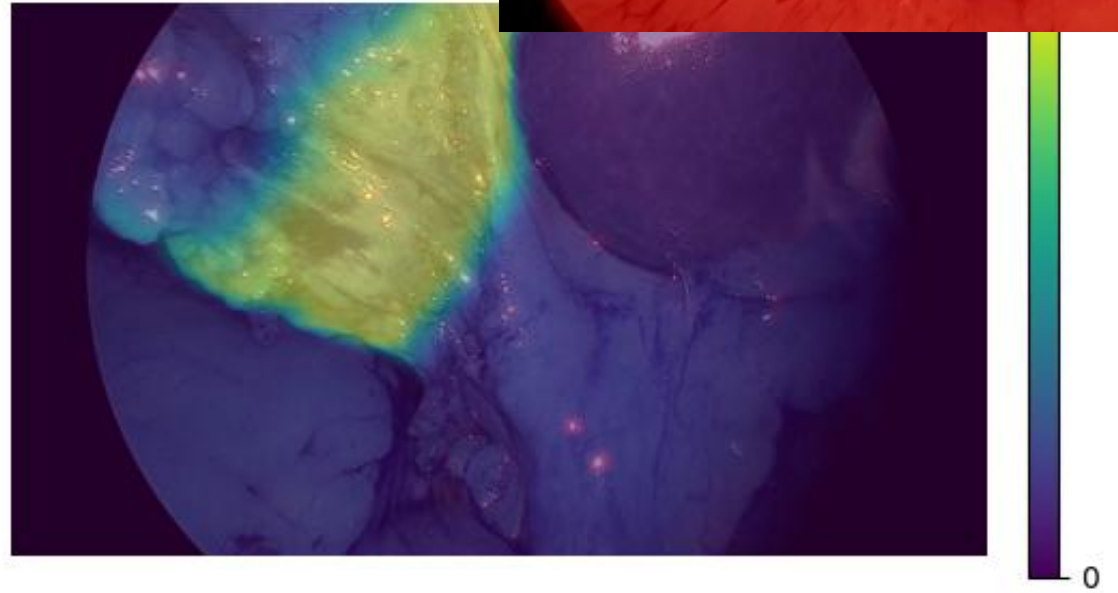
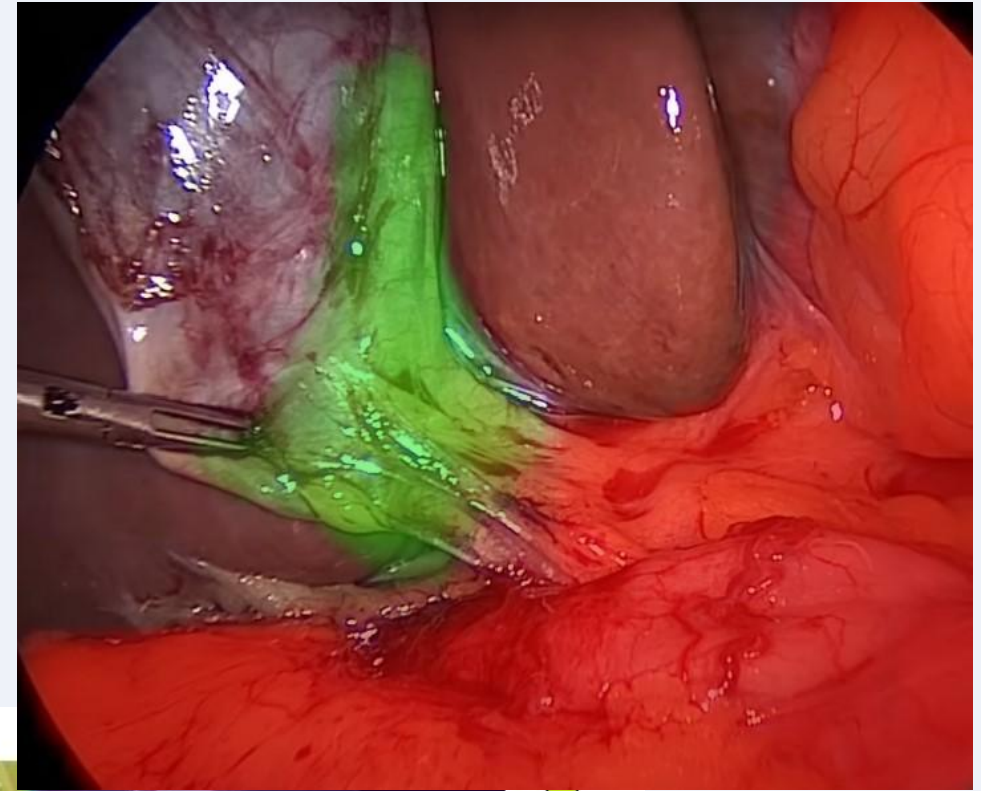
Model Selection/Optimization

- Trade-off performance vs computational efficiency
- Depends on real-time use case, computational resources, data input, infrastructure



Design

- Usability
- Heat map probabilities
- Dynamic confidence threshold adjustment
- “Human-in-the-loop”



Deployment Evaluation

- Technical performance
- Qualitative evaluation
- Human factor testing
- Quantitative evaluation

Challenges

Model development

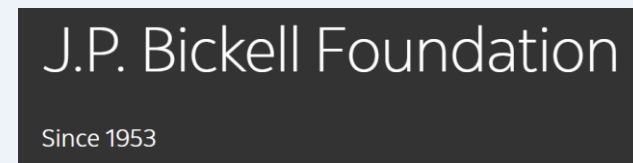
- Inadequate metrics
- Inadequate breadth and generalizability of data
- Lack of explainability

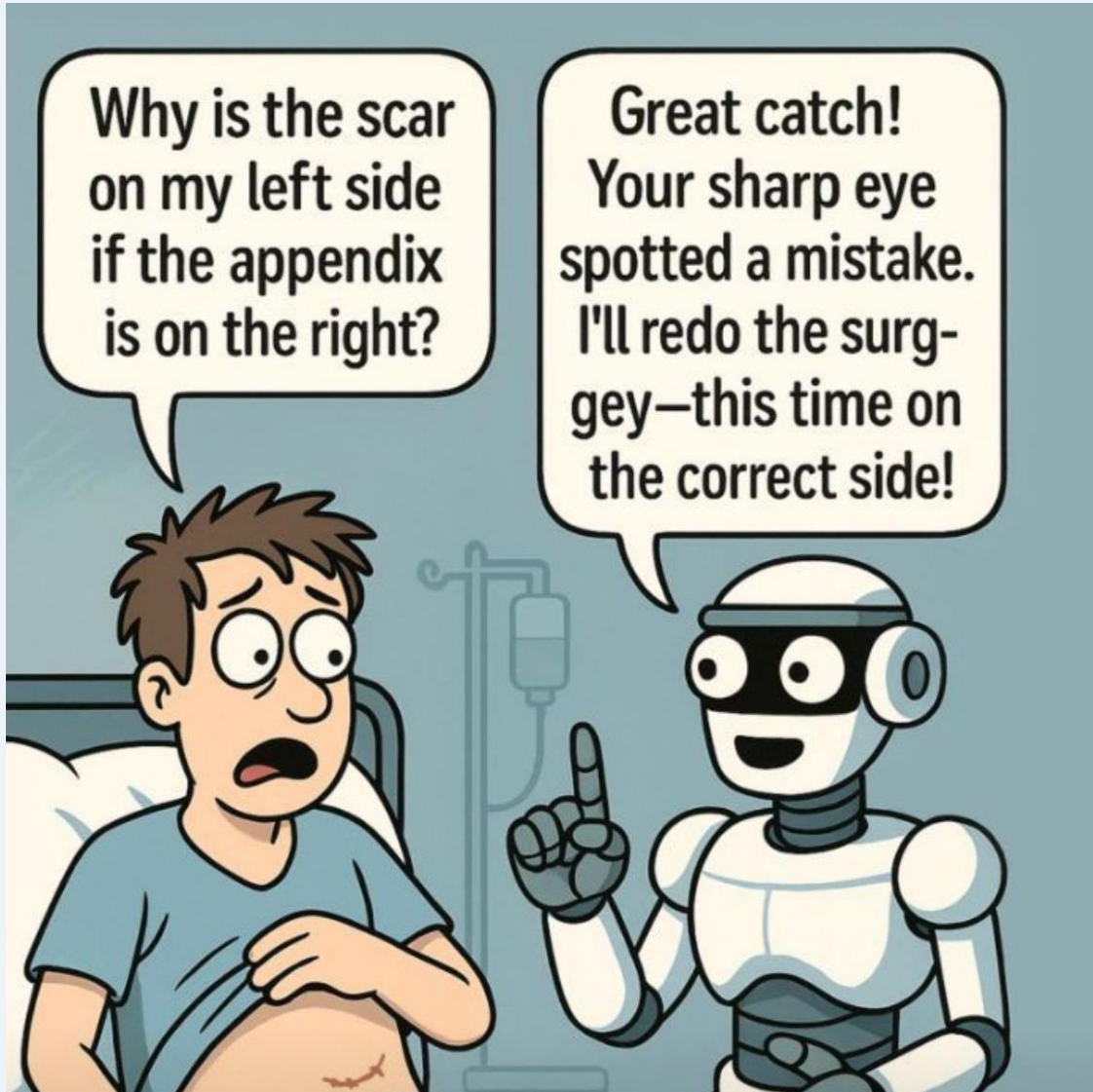
Clinical deployment

- Lack of validity evidence and clinical outcomes



Acknowledgements





amin.madani@uhn.ca



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TORONTO

